Transformer-Based Fake News Detection

MAJOR REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

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V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

Approved by AICTE &Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
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Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this project report titled **"TRANSFORMER - BASED FAKE NEWS DETECTION"** is a bonafide record of work done by **Varalakshmi Maddireddy (228W1A1233) and Lavanya Ravilla (228W1A1255)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	Transformer-Based Fake News Detection
2	Student Names & Numbers	Varalakshmi Maddireddy(228W1A1233) Lavanya Ravilla(228W1A1255)
3	Name of The Guide	Dr. M. Suneetha, Dean R&D, Professor &Head
4	Research Group	Deep Learning
5	Application Area	News and Media Systems
6	Aim of the Project	To develop a transformer-based automated system using BERT and DistilBERT that accurately classifies news content as real or fake by capturing deep contextual relationships, while improving efficiency and interpretability.
7	Project Outcomes	. Implemented advanced text preprocessing and WordPiece tokenization . Developed and evaluated BERT and DistilBERT models for fake news classification with improved accuracy and efficiency.

Student Signatures

1.

2.

Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our Principal **Dr. A.V. RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report , I would like to extend my honor to **Dr. M. Suneetha**, **Dean R&D, Professor & Head of the Department**, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr. M. Suneetha, Ph. D., Dean R&D, Professor & Head of department of the Department, IT** for **her** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficult on my part to finish the project work.

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ABSTRACT

The rapid growth of social media is a big part of how information spreads these days, but it has also made it very easy and hard to stop the spread of false information. It takes a long time to check facts the old-fashioned way, and it's not good for dealing with all the content online. Older deep learning models have some problems, especially when it comes to understanding long and complicated news articles. A transformer-based models is used for efficient false news identification in order to get over these restrictions. To improve contextual comprehension and semantic representation of textual material, transformer models like DistilBERT and Bidirectional Encoder Representations from Transformers (BERT) are used. Text cleaning and tokenization are used to preprocess a dataset of authentic and false news articles. Word Piece encoding is then used to translate the text into useful input representations for transformer models. We use performance metrics like accuracy, precision, recall, and F1-score to train and test the models. Experimental results show that BERT is more accurate because it understands context in both directions deeply. DistilBERT, on the other hand, gives similar results with less computational complexity and faster inference. These models demonstrate improved effectiveness over traditional deep learning approaches in handling complex linguistic patterns. The system is further capable of providing real-time predictions, allowing users to analyze news content and determine its authenticity instantly. The proposed approach shows strong potential for integration into digital platforms, contributing to the reduction of misinformation and supporting reliable information dissemination.

Keywords: Fake News Detection, Transformer Models, BERT, DistilBERT, Natural Language Processing, Text Classification, WordPiece Tokenization.

CHAPTER-1

INTRODUCTION

This chapter introduces the foundation and motivation behind the project, provides the basic definitions and background knowledge necessary to understand the work, outlines the problem statement, discusses the societal impact of the proposed work, and concludes with a summary.

1.1 Origin of the Problem

The rapid growth of social and digital media platforms has brought about some considerable changes concerning how information has been created and consumed in today's world. In addition to giving people fast access to information, the new technology has brought along with it many challenges concerning the proliferation of misinformation. Contrary to traditional media, the new media platforms lack an efficient system for verifying content that is uploaded on such platforms. As such, fake news has managed to proliferate very fast to reach many individuals within a relatively shorter period. The biggest challenge that has come out of this trend is the lack of efficiency in information verification systems. Traditional fact checking is largely manual; hence, it lacks efficiency to keep up with the pace of information sharing on digital media platforms [1].

Deep learning models improved on pattern and relationship analysis, and consequently, they outdid the former in efficiency and effectiveness. Recent models have continued to advance the process of document understanding and text classification. These approaches are capable of capturing deeper contextual relationships and handling complex textual structures in news content. As the size and complexity of digital information continue to grow, ensuring accurate analysis and maintaining meaningful interpretation of content remain important challenges. These challenges highlight the need for an efficient and reliable system that can effectively process large-scale textual data and accurately classify information. Therefore, this project focuses on developing an approach that can analyze news content, capture important patterns and contextual relationships, and provide reliable classification of news as real or fake.

1.2 Basic definitions and Background

1.2.1 Fake News Detection

Detection of fake news involves checking whether the news being disseminated on a news website has real or fabricated content. As communication becomes increasingly digital, fake news spreads rapidly across all available channels on social media, rendering manual verification impractical. Computer systems become useful in this context to help analyze and categorize the text data as either fake or authentic news [1].

1.2.2 Natural Language Processing (NLP)

A natural language process (NLP) is an artificial intelligence technology used by computers to comprehend and manipulate human language. NLP has been useful in analyzing articles of news using different linguistic information including semantic and contextual analysis. Early steps in our implementation process of analyzing the articles used NLP techniques such as tokenization, text cleaning, and TF-IDF vectorization [2].

1.2.3 Machine Learning Approaches

Machine Learning techniques are used to classify news articles based on patterns learned from data. Algorithms such as Naïve Bayes, Logistic Regression, and XGBoost analyze features extracted from textual data to predict whether news is real or fake. In the initial phase of this project, models like XGBoost demonstrated strong performance by capturing complex patterns in data and achieving higher accuracy compared to simpler models [3].

1.2.4 Deep Learning Approaaches

Deep learning models further improved fake news detection by capturing sequential and contextual relationships in text. LSTM models learn long-term dependencies, while BiLSTM processes text in both forward and backward directions to enhance contextual

understanding. Hybrid models such as CNN–BiLSTM combine local feature extraction with sequence learning, resulting in improved accuracy and better handling of complex textual patterns. These models represent an advancement over traditional machine learning techniques in understanding news content [4].

1.2.5 Transformer-Based Models

Transformer-based systems are considered the latest innovation for detecting fake news. The self-attention system employed in BERT and DistilBERT allows the model to incorporate the contextual interaction at a much deeper level in the entire sequence of text. The ability of the model to determine the semantics of a word in relation to its context in the phrase makes these models unique from other previous models. Based on the research findings, DistilBERT gave good results with less computational cost; however, BERT performed well owing to its good contextual understanding ability [5].

1.2.6 Text Preprocessing and Feature Representation

Preprocessing of the text is a crucial part of text preparation. It includes such operations as noise removal, text cleaning, sentence tokenization, and word-to-number transformation. Traditionally, more primitive algorithms, such as TF-IDF, were used, but now more sophisticated approaches like WordPiece tokenization are being implemented in complex models. Appropriate preprocessing guarantees enhanced accuracy of classification results [1].

1.2.7 Evaluation Metrics

These measures of evaluation help in making accurate classifications regarding the outcomes of the false news detection systems and evaluating their efficiency. The measures of evaluation utilized in determining the efficiency of false news detector systems include accuracy, precision, recall, and F1-score. Precision and recall help in measuring the efficiency of predictions made, while accuracy helps in measuring the efficiency of classification.

The F1-score measures the efficiency of the model. Confusion matrix is another visualization measure that helps in showing classification results. Transformer-based models perform better compared to other models based on the experimental results of this study.

Background

With the rapid development in the field of digital media and online portals has led to the emergence of the problem of fake news dissemination. Fake news may affect public perception, cause misunderstanding, and adversely affect decision-making processes in political, healthcare, and social domains. Conventional techniques employed to validate news articles involved manually checking facts, but due to their time-consuming nature, they are unable to handle the vast amount of data being created daily. The initial machine learning techniques used for the detection of fake news only considered simple textual and statistical characteristics and had little capability in grasping the semantic meanings of texts. The emergence of the field of deep learning and neural networks has enhanced the capability of processing textual data by considering their sequential and semantic connections. Nevertheless, these models have been unable to capture the long-term dependencies and rich contextual meanings of texts. To solve this problem, transformer-based models have been developed [2].

This work leverages sophisticated models like BERT and DistilBERT to process news articles and identify if they are genuine or not. These algorithms make use of the mechanism of self-attention to understand the contextual relation of words within the article. As a result, this makes it easier to understand the context of an entire article [5]. Unlike the previous work, the system is based on features that take into account semantics and contextual information. In this proposed model, textual information is processed and useful features are extracted using transformer-based embeddings to classify real and fake news articles. Moreover, an easy-to-use interface may also be created that enables users to input news articles and receive real-time predictions.

1.3 Problem Statement

The quick increase in the dissemination of fake news on online platforms has proved to be a major problem and cast doubt on the credibility and integrity of information online. As more and more textual data is generated every day, it becomes only viable to validate it manually, so more automated systems of detecting fake news become imperative.

Conventional use of machine learning mostly uses superficial characteristics of language, including word frequency, syntax. But, such techniques are frequently not able to reflect more semantic relations in complex news stories. Although the contextual understanding has improved with the deep learning techniques, there are still constraints associated with effective applications on long sequences of text using the techniques. Moreover, differences in writing styles, changes in misinformation strategies and dissimilar content structures are all complicative to a classification task. The other serious issue is a difficulty in maintaining an accuracy and computational efficiency in handling large-scale datasets [2].

In an attempt to overcome these shortcomings, this project has suggested a transformer-based solution with fake news detection through Bidirectional Encoder Representations with Transformer (BERT) models and DistilBERT models. Such models are geared towards recognizing contextual dependencies in the full sequences of text, which they can be used to obtain a more detailed sense of semantic meaning. DistilBERT is a lightweight one, which also provides speed of inference with competitive accuracy [5].

The key goal of the proposed system is to achieve better classification results with the use of contextual embeddings that will serve to propose a scalable and efficient system to detect fake news.

1.3.1 Objectives

- **Development of Transformer-Based Models**

  For creating and developing sophisticated transformation models like BERT and DistilBERT for the purpose of detecting fake news by leveraging deep contextual and semantic connections within the text.

- **Comparative Performance Assessment**

  For assessing the performance of the transformation models through measurements like accuracy, precision, recall, and F1 score and analyzing their efficiency and complexity.

- **Handling Textual Data Challenges**

  For overcoming problems like noisy data, differences in writing styles, and extensive textual data by utilizing efficient preprocessing and tokenization methods.

- **Real-time Fake News Classification**

  For designing a system that can effectively distinguish between real and fake news articles in real-time and provide timely predictions.

1.3.2 Outcomes

- **Accurate Fake News Classification**

  The model can classify news articles as either real or fake based on its ability to learn complex context and semantics using transformer-based architectures.

- **Better Model Predictions**

  Using more sophisticated models like BERT and DistilBERT ensures superior performance over conventional machine learning or deep learning models, particularly with respect to context prediction.

- **Efficient and Scalable System**

  This model enables rapid processing of vast amounts of text data, while the speedier DistilBERT ensures quick predictions with lower computational requirements.

- **Reliable and Interpretable Results**

  The model produces predictable and easily interpretable results, thereby building user confidence in the model, which proves to be handy in applications such as news validation and content filtering.

1.4 Societal Impact of Proposed Work

- The proposed fake news detection model based on transformers is useful in minimizing the proliferation of false information through automatic classification of news as authentic or fake, thereby enhancing the authenticity of information available on the internet [1].
- The detection of fake news assists in better decision-making regarding important aspects like health care, governance, and even in social media platforms through accurate interpretation of the news content. Early detection of fake news is crucial in avoiding confusion and taking wrong decisions [4].
- Automatic detection of fake news will minimize the need for manual checking of facts, and thereby make the process more efficient, effective, and faster.

1.5 Applications

- Social Media Monitoring: This system could be implemented on social media sites for automatically detecting and marking fake news articles in order to prevent the dissemination of such misinformation [1].
- News Verification System: This could help the media outlets verify the authenticity of news stories prior to publishing them in order to ensure their credibility.
- Content Moderation & Cybersecurity: The system could be used to identify any misleading or threatening material that could be posted on websites.
- Search engine optimization and filtering: The system could also be incorporated with search engines where more credible news would be ranked higher whereas misleading information and fake information would be minimized when a result is searched.

This chapter elaborates on the background and justification for the fake news detection system being proposed. It covers the difficulties encountered when trying to detect misinformation within massive data sets that are mostly made up of text. It has further delved into different existing approaches ranging from those based on machine learning algorithms all the way to those based on deep learning. Its significance is highlighted in terms of how it contributes towards enhancing the efficiency of fake news detection systems. It was also covered why more advanced techniques are required in order to be able to capture the deeper semantics of news data.

CHAPTER-2
REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and considered all the points mentioned in the papers.

2.1 Description of Existing Systems

A Joint Learning Framework for Fake News Detection

Authors: Muhammad Abdullah, Hongying Z., Javed A., Mamyrbayev O., Caraffini F., Eshkik

Year of Publishing: 2025

Observations:

This study proposes a joint learning framework for fake news detection using the FakeNewsNet repository, which includes Politifact and Gossip Cop datasets. The model utilizes an enhanced BERT-based transformer approach integrated with joint learning techniques such as Named Entity Recognition, Random Forest Classification, and stance detection. It also incorporates hierarchical attention and task-specific masking to improve contextual understanding. The experimental results show that the model achieves state-of-the-art accuracy on the Gossip Cop dataset. The study highlights strong contextual learning capabilities; however, it also notes that the model involves high computational cost and a complex architecture [1].

Fake News Detection: Comparative Evaluation of BERT-like Models and Large Language Models with Generative AI-Annotated Data

Authors: S. Raza, D. Paulen-Patterson, C. Ding

Year of Publishing: 2025

Observations:

This study performs a comparative evaluation of BERT-like models and large language models using a custom dataset of news articles annotated with GPT-4 and verified by human experts. The authors utilize fine-tuned BERT encoder models along with autoregressive large language models, including instruction-tuned LLMs with majority voting during inference. The results

show that BERT-like models generally achieve higher classification accuracy, while large language models demonstrate better robustness to variations in text. The study also highlights that combining AI-generated labels with human verification improves overall model performance [2].

A Knowledge-Based Deep Learning Approach for Automatic Fake News Detection using BERT on Twitter

Authors: Vinita Nair, Jyoti Pareek, Sanskruti Bhatt
Year of Publishing: 2024

Observations:
This research proposes a knowledge-based deep learning framework for fake news detection using BERT on Twitter data. The dataset includes information enhanced with knowledge base features such as SPO triples, sentiment polarity, occurrence, and topic modelling. The system integrates a knowledge-based pipeline combining Named Entity Recognition and Latent Dirichlet Allocation with deep learning models. The results indicate that knowledge-based features significantly improve pattern discovery and classification performance, while BERT demonstrates strong effectiveness in handling contextual relationships. However, the approach depends on knowledge base construction and is influenced by noise in social media data [3].

Enhancing Fake News Detection Using BERT: A Comparative Analysis of Logistic Regression, Random Forest, LSTM and BERT

Authors: A. Ramzan, R.H. Ali, N. Ali, A. Khan
Year of Publishing: 2024
Observations:
This study compares traditional machine learning models such as Logistic Regression and Random Forest, along with deep learning models like LSTM, against transformer-based BERT for fake news detection. The experiments are conducted on a relatively small real-world dataset to simulate practical conditions. The results show that BERT outperforms all other models in terms of accuracy and contextual understanding. The study highlights that transformer-based

models are more effective in capturing semantic relationships even with limited data, though dataset size may influence generalization [4].

Detection of Turkish Fake News from Tweets with BERT Models

Authors: G.K. Koru, Ç. Uluyol

Year of Publishing: 2024

Observations:

This work focuses on detecting fake news from Turkish tweets using fine-tuned BERT models. The dataset consists of both human-annotated and machine-generated samples in the Turkish language. The study demonstrates that BERT models achieve higher accuracy compared to traditional machine learning approaches in multilingual and low-resource settings. It also shows that transformer models are effective in capturing language-specific patterns. However, the approach is limited to a specific domain, which may affect its generalization across different datasets [5].

Word2Vec and LSTM based deep learning technique for context-free fake news detection

Authors: Abhishek Mallik; Sanjay Kumar

Year of Publishing: 2024

Observations:

This study developed an advanced LSTM-based approach for classifying fake and real news articles using refined preprocessing and improved sequential learning techniques. The authors emphasized that capturing long-term dependencies in narrative structures is crucial for identifying fabricated information. Their framework incorporated enhanced tokenization, embedding optimization, and stacked LSTM layers to extract deeper semantic patterns. Experimental evaluations showed notable improvements in accuracy and robustness compared to conventional LSTM models. The findings confirmed that sequential deep learning remains one of the most reliable approaches for detecting misinformation across diverse textual datasets. Moreover, the research underscores the need for integrating the linguistic context in combination with deep learning in order to enhance the accuracy of detection. This ability of the

proposed method to adjust to changes in the style of the text renders it highly practical in the detection of fake news [6].

A novel approach to fake news classification using LSTM-based deep learning models

Authors: Halyna Padalko

Year of Publishing: 2024

Observations:

This study developed an advanced LSTM-based approach for classifying fake and real news articles using refined preprocessing and improved sequential learning techniques. The authors emphasized that capturing long-term dependencies in narrative structures is crucial for identifying fabricated information. Their framework incorporated enhanced tokenization, embedding optimization, and stacked LSTM layers to extract deeper semantic patterns. Experimental evaluations showed notable improvements in accuracy and robustness compared to conventional LSTM models. The findings confirmed that sequential deep learning remains the most reliable approaches for detecting misinformation across diverse textual datasets [7].

Fake News Detection Using Dual BERT Deep Neural Networks

Authors: Mahmood Farokhian, Vahid Rafe, Hadi Veisi

Year of Publishing: 2024

Observations:

This study proposes a dual BERT-based deep neural network model for fake news detection by combining both textual and visual information. The approach utilizes a multi-modal transformer framework where textual features are extracted using BERT, while visual features are captured through a two-level visual feature extraction mechanism. By integrating both image and text data, the model enhances the ability to detect misleading content that may not be identifiable through text alone. The experimental results demonstrate improved classification performance and better contextual understanding compared to traditional text-only approaches. The study highlights the effectiveness of multi-modal learning in improving fake news detection accuracy, especially in complex real-world scenarios [8].

Fake news detection via NLP is vulnerable to adversarial attacks

Authors: Z. Zhou, H. Guan, M. M. Bhat, J. Hsu

Year of Publishing: 2023

Observations:

This study proposes the weaknesses of NLP-based methods for identifying fake news upon facing an attack. The writers have explained that most of the current approaches depend on linguistic analysis but not on verifying facts, which is why they can be easily fooled by an attacker. In particular, the study has shown that attackers can modify either factual details or writing styles to make the detectors predict wrong results. This effect could be seen in the experiments carried out using the Fake box system. In order to resolve this issue, the study suggests including techniques of fact verification along with linguistic analysis. Also, the creation of crowdsourced knowledge graphs will prove helpful. They also propose the use of a crowdsourced knowledge graph to collect timely and accurate factual information, which can strengthen model performance [9].

Detection of fake news using machine learning and NLP algorithms

Authors: N. N. Prachi, M. Habibullah, M. E. H. Rafi, E. Alam, R. Kha

Year of Publishing: 2023

Observations:

This This study deals with the problem of recognizing fake news in the age of rapid growth of information available via online sources, particularly social media platforms, where information is distributed extensively at an accelerated pace. In other words, it becomes increasingly complicated to distinguish between reliable and fake data because of the fast development of technology used to distribute vast amounts of information without applying adequate verification procedures. Therefore, the present paper discusses the possibility of applying machine learning and natural language processing techniques to detect fake news efficiently. It should be emphasized that logistic regression, decision trees, Naive Bayes, support vector machines, long short-term memory, and BERT were chosen as classification algorithms in order to train the models to differentiate between genuine and fake news [10].

2.2 Summary of Literature Study:

The summary of literature review in Table 2.1 presents the literature reviewed for the fake news detection system demonstrate that some researches aimed to develop and evaluate methods for enhancing fake news detection through the use of transformer models such as BERT. Some authors proposed joint learning framework combining different methodologies such as NER, stance detection, and classification techniques in order to increase contextual knowledge and ensure higher accuracy.

Moreover, there were some papers comparing BERT-like models and language models that demonstrated higher classification accuracy and robustness of the former models. Moreover, some researches proposed knowledge-based approach, which was supposed to use extra attributes alongside deep learning models in order to improve pattern detection capability especially in social media data. Furthermore, there are studies that compared machine learning models, deep learning models, and transformer models and demonstrated superiority of the latter ones in terms of semantic relationship identification capability [5].

Moreover, some researches dealt with fake news detection for particular languages through BERT models, highlighting their performance even when applied to multilingual and low resource settings. Some researchers also made use of deep learning through LSTMs in order to consider sequences and long-term relationships within texts and to increase classification precision. Moreover, one can state that the research related to fake news detection demonstrates an evolution from machine learning approaches to deep learning and transformer models [6].

The significance of context-awareness, feature integration, and efficiency of transformers in machine learning algorithms was shown in these studies, proving the application of transformers in the proposed system.

Table 2.1: Literature Review

S. No	Title& Publication Details	Datasets	Algorithms/ Technologies Used	Purpose	Remarks
1	A Joint Learning Framework for Fake News Detection (Muhammad Abdullah et al., 2025, Displays – Elsevier)	FakeNewsNet (PolitiFact, GossipCop)	Enhanced BERT, Joint Learning, Hierarchical Attention	To improve fake news detection using combined learning techniques	strong contextual understanding; high computational cost
2	Fake News Detection: Comparative Evaluation of BERT-like Models and Large Language Models (S. Raza et al., 2025, Knowledge and Information Systems)	AI-annotated custom dataset (GPT-4 + human verified)	Fine-tuned BERT models, LLMs, Majority voting	To compare performance of BERT and LLM models	LLMs more robust; AI + human labels improve performance
3	A Knowledge-Based Deep Learning Approach for Fake News Detection using BERT on Twitter (Vinita Nair et al., 2024, Procedia Computer Science)	Twitter dataset with knowledge base features	BERT + Knowledge-based methods	To improve detection using external knowledge features	Improves pattern detection; strong performance; depends on knowledge base and noisy data
4	Enhancing Fake News Detection Using BERT: A Comparative Analysis (A. Ramzan et al., 2024, IEEE)	Small real-world dataset	Logistic Regression, Random Forest, LSTM, BERT	To compare ML, DL, and Transformer models	BERT outperforms all; better contextual understanding; dataset size affects generalization

5	Detection of Turkish Fake News from Tweets with BERT Models (G.K. Koru et al., 2024, IEEE Access)	Turkish Twitter dataset	Fine-tuned BERT	To detect fake news in multilingual setting	Higher accuracy than ML; works well in low-resource languages; domain-specific limitation
6	Word2Vec and LSTM based Deep Learning Technique for Fake News Detection (Abhishek Mallik et al., 2024)	Real-world news datasets	Word2Vec, LSTM	To classify fake news using sequential learning	Improved accuracy; captures long-term dependencies; better than basic models
7	A Novel Approach to Fake News Classification using LSTM-based Models (Halyna Padalko, 2024)	News datasets	LSTM, Embedding techniques	To enhance classification using deep learning	Improved robustness and accuracy; effective for sequential data
8	Fake News Detection Using Dual BERT Deep Neural Networks (Farokhian et al., 2024, Springer)	Multi-modal dataset (text + images)	Dual BERT, Multi-modal transformer	To detect fake news using text and visual features	Improved performance; better contextual understanding; effective for real-world scenarios
9	Fake news detection via NLP is vulnerable to adversarial attacks (Zhou et al., 2019, arXiv)	Common benchmark datasets (e.g., news datasets used in NLP studies; adversarially modified text samples)	Machine Learning models, Natural Language Processing (NLP), Adversarial Attack Techniques.	The reason is to explore how convincing adversarial text manipulations are to fake news detection models.	Exposes significant weakness of the NLP-based systems: prone to adversarial attacks; more robust models are required.

10	Detection of fake news using machine learning and natural language processing algorithms (Prachi et al., 2022, Journal of Advances in Information Technology)	Fake news datasets that are publicly available (e.g. Kaggle datasets, news corpora).	Machine Learning (Naive Bayes, SVM), NLP techniques (TF-IDF, preprocessing)	To create a fake news detection system based on traditional ML and NLP systems.	Delivers baseline performance; weak at a deep semantic context relative to models based on transformers.

This paragraph gives the full discussion of the literature in terms of fake news detection whereby the development of the methods employed to solve this issue is discussed. It talks about different models used in prior research among which are discrete machine learning models, deep learning models, along with newer models based on transformers. Earlier algorithms mostly used feature engineering techniques like term frequency-inverse document frequency (TF-IDF) and classification models like Naive Bayes and Support Vector Machines that achieved a reasonable level of performance, but lacked the ability to learn more nuanced semantic dependencies in the text.

As deep learning progressed, other models like recurrent neural networks (RNNs) and long short-term memory (LSTM) networks emerged to learn more about the contextual dependencies of textual information. Nevertheless, long sequences may prove problematic in these models and significant computational resources were in need. More recently, methods that use transformers (such as BERT and its variants) have proven to be superior, as they manage to capture both the forward and reverse context as well as semantic meaning at document scale.

Among the most notable enhancements made by such advanced models, the literature also mentions enhanced accuracy, enhanced contextual understanding, and scalability. Meanwhile, it identifies a number of drawbacks, including susceptibility to adversarial attacks, inability to effectively run very large datasets, and issues with model interpretability. Besides, differences in writing patterns, changing patterns of misinformation, and differences specific to domains are still challenging to current systems.

CHAPTER-3
PROPOSED METHODOLOGY

The proposed methodology presents a complete workflow for detecting fake news using sophisticated transformer-based methods. It starts from data collection and processing, feature extraction, and model creation, followed by the implementation of machine learning algorithms such as BERT and DistilBERT. The developed system will be able to process texts efficiently since it can analyze texts through the identification of their contextual dependencies. The text preprocessing stages such as text cleaning, tokenization, and encoding will allow transforming raw text data into features. The methodology is structured to maintain high accuracy by capturing both contextual and sequential patterns in textual data, as demonstrated in recent research.

3.1 System Architecture Diagram

This Figure 3.1 illustrates the architecture diagram above illustrates the entire flow of the suggested transformer-based solution for fake news detection. The flow starts from the input data that is a set of news articles that are labeled whether they are real or fake. As mentioned above, the unprocessed texts undergo preprocessing procedures during which all the punctuation, and noises are removed from the text. Afterward, the text is split into tokens so that the data can be transformed into vectors that will be fed into models for further processing. Once the text undergoes preprocessing, the next step is to convert it into numbers or vectors with the help of various embedding algorithms. In our case, WordPiece tokenization is performed in order to create embeddings that are capable of reflecting the semantic content of the words contained in the text. Finally, once the preprocessing stage is completed, the input goes to the transformer-based models, namely, BERT and DistilBERT. These models comprise several layers in which self-attention mechanisms are applied so that the relative significance of individual words in the sentence is identified [7].The transformer model converts the inputs using different encoding layers that determine the attention weights for the relevant segments of text. In the process, the model creates a representation of the full news article derived from a unique token indicating the overall sense of the document. This representation is then passed to a classification layer, where the model predicts whether the news article is

real or fake. The final output is generated based on probability scores. According to the results, BERT achieves higher accuracy due to its deeper architecture, while DistilBERT provides faster predictions with lower computational cost, making the system both effective and efficient.

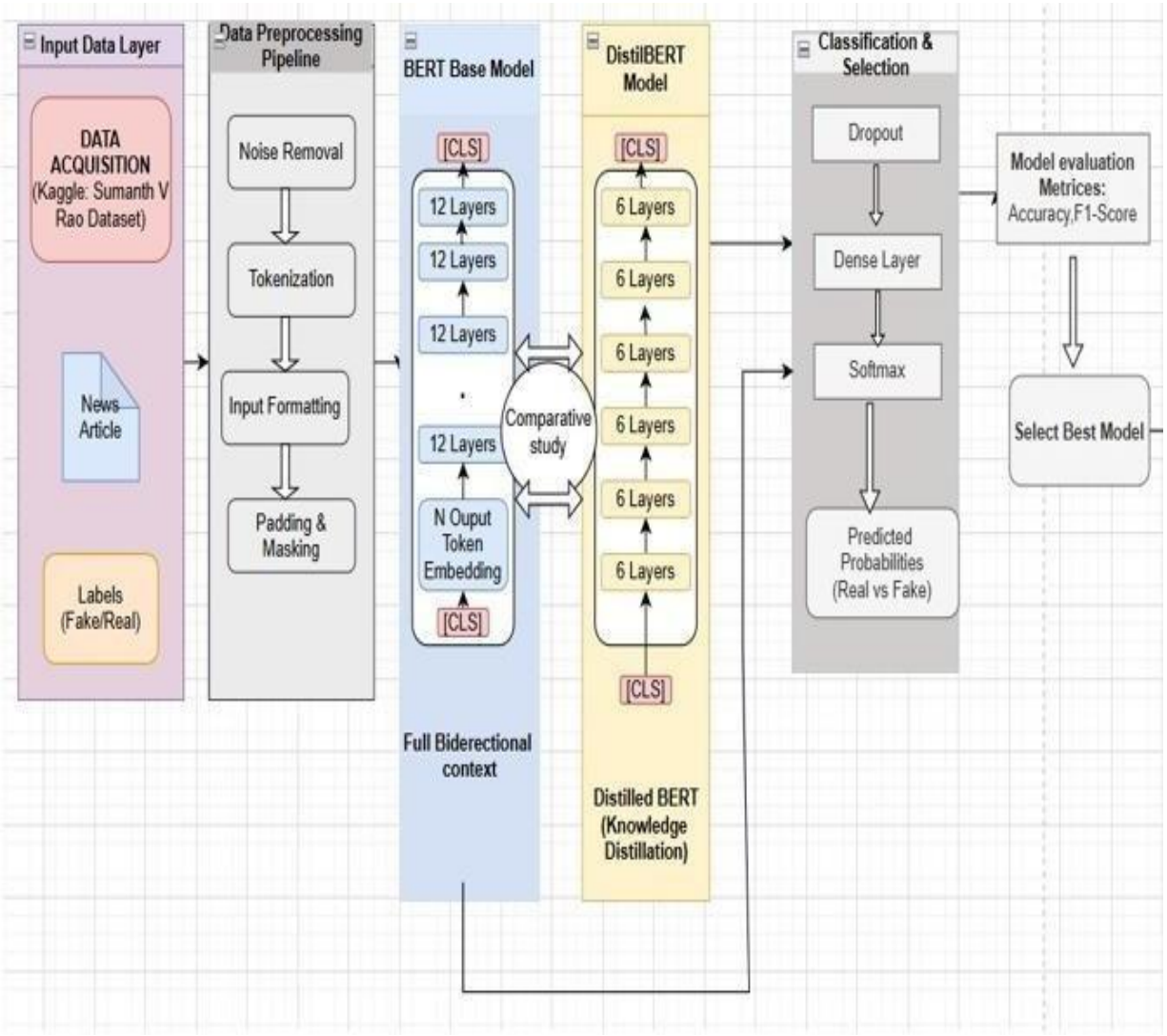


Figure 3.1. Architecture Diagram of Fake News Detection

After preprocessing, the system uses transformer-based embedding that transforms tokens into contextual vectors. This allows the system to not only extract the meaning of words but also their relations with one another in the sentence. The extracted embeddings are then fed into transformer models such as BERT and DistilBERT based on the model architectures.

- BERT (Bidirectional Encoder Representations from Transformers) analyzes the whole text in both directions, making the system capable of capturing the context of each word relative to other words. To determine the words that are essential in a sentence, BERT makes use of numerous transformers with self-attention mechanism to detect them. There is a specific token referred to as [CLS] that captures the overall meaning of the text. Afterward, the meaning is sent through a classifier for determining the authenticity of the news.
- DistilBERT is a smaller and efficient variant of BERT where almost all the capabilities are retained along with fewer layers and parameters used. Similar to BERT, it too uses transformer model and self-attention for understanding context, but in a more computationally efficient manner. DistilBERT takes less time in processing the inputs and needs minimal memory, making it suitable for real-time applications as well. Its output representation will be similarly sent to a classifier layer to make final predictions.
- Dense Layer (Fully Connected Layer) performs analysis on the features obtained through the transformation models in order to learn the patterns that are needed for making a classification. The input of this layer is the context representation that comes from [CLS] token. With learned parameters, the representation is transformed into something more predictive. This step is crucial for consolidating all important information that will be used further to classify the texts.
- Softmax Function is a function used in the final part of classification to produce probabilities. Namely, softmax outputs probabilities where each element represents a probability of belonging to class (in our case, real/fake). So, summing up all the probabilities gives 1, which means that the predicted text belongs to one of these classes with a certain probability. The class that got the highest probability is selected as the result.

3.2 Description of Datasets and Requirement Specification

3.2.1 Dataset Description

Purpose:

The dataset is used to train and evaluate the proposed transformer-based fake news detection system. It contains labeled news articles that are classified as fake (0) or real (1). The dataset consists of approximately 44,898 news articles, collected from publicly available sources. It is designed to help the model learn patterns and relationships in textual data for accurate classification.

Each record in the dataset includes the following components:

- Title: Represents the headline of the news article, which provides a short summary of the content.
- Text: Contains the complete news article, which is used as the primary input for the model.
- Label: Indicates whether the news is fake (0) or real (1), used as the target for classification.

The dataset is divided into training, validation, and testing sets to ensure proper model learning and evaluation.

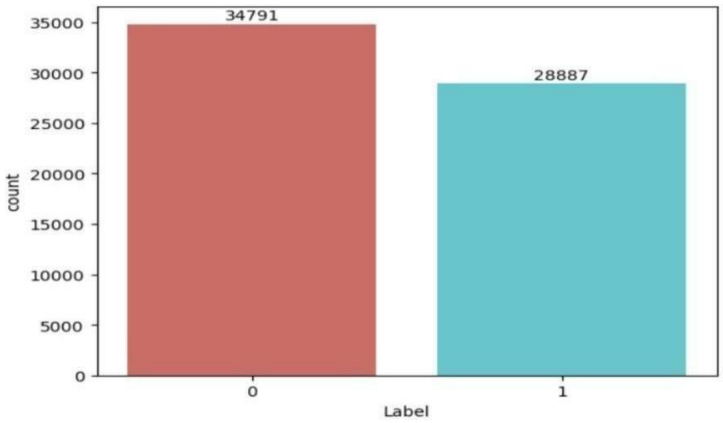


Figure 3.2. Label Frequency Distribution

Source and Availability:

- The dataset is publicly available on Kaggle (Fake News Classification Dataset).
- It is structured in CSV format, making it easy to load and preprocess using Python libraries like Pandas.

Preprocessing Requirements:

- **Text Cleaning:** Removal of punctuation, special characters, and unnecessary symbols to improve data quality. This step ensures that the input text is free from noise and irrelevant elements that may affect model performance. Text cleaning also facilitates the standardization of data, making it easier for the model to understand. Moreover, by cleaning the text, we can avoid problems like inconsistent spacing, formatting, and other unnecessary patterns in the data. Text cleaning also allows us to ensure consistency in all texts, which helps in learning by the model. With this process, irrelevant data is also eliminated, allowing the model to focus only on relevant information.

- **Tokenization:** Tokenizing text through WordPiece tokenization for transformer models. Through tokenizing text, the sentences are split up into subword pieces that make it possible for the model to be able to understand words better and also be able to handle any rare or unknown words that may appear in the text. Tokenization also plays an important role in helping the model understand how words interact in a sentence as well as the sentence structure. The use of subwords allows the model to be able to capture meanings even in cases where words have never been seen before.

- **Stop Word Handling:** Reducing less important words while preserving meaningful context. Rather than eliminating all stop words from the text, some of the key contextual words are kept in place to ensure that the meaning of the sentence is not distorted, especially when using transformer models. The technique enables the retention of natural sentence flow without the need for excessive data noise. Furthermore, the elimination process allows the machine learning model to concentrate on key words without neglecting

the importance of connector words that define the sentence's structural components. Selective stop word removal optimizes the data simplification-contextual relationship equilibrium.

- **Encoding:** Tokenization and transformation into numeric form for the BERT and DistilBERT models. This involves transforming words into numbers that can be used in the model. This allows the machine learning system to make sense of the inputs provided. The model uses tokenization to understand what words mean as well as their relationship with other words within the same sentence. Further, encoding allows the machine learning system to understand context because similar representations will be provided for words that have the same meaning. Therefore, this becomes an important part of training the machine learning system to become better at predicting various outcomes [8].

3.2.2. Software Requirements

Programming Language:

- Python: For implementation of full fake news detection model. Python is a powerful language which offers flexibility and ease of use when it comes to coding. Also, python has vast capabilities for working with data including data analysis, machine learning and deep learning.

Libraries:

- Data Handling: Pandas, NumPy for loading and processing datasets.
- NLP Processing: NLTK, spaCy for text preprocessing.
- Transformer Models: HuggingFace Transformers for implementing BERT and DistilBERT.
- Deep Learning Framework: PyTorch for model training and evaluation.
- Visualization: Matplotlib, Seaborn for plotting graphs and confusion matrices.
- Deployment: Flask for building a web-based application.

3.2.3. Hardware Requirements

- Processor: Multi-Core Processor (i5/i7 Intel or Ryzen 5/7 AMD), which processes the data for pre-processing and any computational processes. More cores help execute the task much faster than otherwise would be possible.

- RAM: The minimum requirement is 8 GB (preferably 16 GB), which ensures that the large datasets and their processing will run smoothly without any issues.

- Storage: Minimum storage space required is 1 GB, which allows one to store all the datasets, trained models, etc., and using SSD storage will speed up the process.

- GPU: Dedicated GPU which accelerates the training and inference of deep learning transformer models such as BERT and DistilBERT.

3.2.4. Development Environment

- Development Tools: For coding and training of the machine learning model, tools such as Visual Studio Code or Google Colab can be used. Both of them have the required functionality for programming and provide an effective working environment.
- Data Storage: Data will be stored in CSV format in order to facilitate the process of handling data, transferring it from one computer to another, and using it with other applications. The simplicity of this file format will also improve its processing.
- Libraries Integration: In Python, there are a number of useful libraries that will allow performing required preprocessing of data. These include Pandas, NLTK, and others.
- Execution Environment: For training and testing of the models, it is important to use the interactive environment that will enable better control of the process.

3.3. Design Methodology

The design methodology of the proposed fake news detection system revolves around the creation of a highly effective and scalable system based on transformer-based machine learning models. It requires significant data processing, ensuring that the system can classify news articles as either real or fake with great accuracy. The process includes different stages, including data collection, preprocessing, feature extraction, model training, and classification. To begin with, the labeled dataset will be gathered and processed. Then, the raw textual data will be preprocessed, turning into a structured form.

Once the preprocessing step is complete, numerical features will be extracted from the textual information, relying on state-of-the-art embedding algorithms. The transformer-based BERT and DistilBERT will be applied, allowing the system to capture the context and meaning of the words depending on other words in their vicinity.While BERT will ensure high precision, DistilBERT will allow for faster execution. Once training is finished, the model will be ready for use and classification of new documents.

3.3.1. Data Collection

The data collection stage involves gathering a comprehensive and well-labeled dataset of news articles that includes both real and fake information. The dataset is obtained from publicly available sources such as Kaggle, which provides a large collection of news data suitable for training deep learning models. Each data entry typically consists of a title, full text content, and a corresponding label indicating whether the news is real or fake. The diversity and quality of the dataset play an important role in improving the performance of the model, as it allows the system to learn various writing styles, topics, and patterns associated with misinformation. To ensure proper model evaluation, the dataset is divided into training, validation, and testing sets. This division helps the model learn effectively from the training data while being tested on unseen data to measure its generalization capability.

The deletion of duplicates and management of missing data can further enhance the quality of the dataset. The use of a balanced dataset containing both genuine and fake news will enable the model to be more accurate and minimize bias. In addition, obtaining the data from various fields like politics, health, and social media will make the model versatile and robust enough to deal with real-world situations.

3.3.2. Pre-Processing

Pre-processing is a crucial step that prepares the raw textual data for effective model training. The collected news articles often contain noise such as punctuation marks, special characters, numbers, and unnecessary symbols, which are removed during the cleaning process. The text is also converted into lowercase to maintain uniformity and avoid duplication of words due to case differences. Tokenization is performed to break down the text into smaller meaningful units using WordPiece tokenization, which is particularly suitable for transformer-based models. This approach helps in handling unknown or rare words by splitting them into subwords. In addition, common but less meaningful words may be handled carefully to maintain important context. These preprocessing steps ensure that the data is clean, structured, and ready for feature extraction, ultimately improving the model's performance.

3.3.3. Feature Extraction

Feature extraction focuses on converting the processed textual data into numerical representations that can be understood by machine learning models. After tokenization, each token is transformed into an embedding vector that represents its meaning. In transformer-based models, these embeddings are not static but contextual, meaning the representation of a word changes depending on its surrounding words in the sentence. This helps the model capture deeper semantic relationships within the text. Special tokens such as [CLS] are used to represent the overall meaning of the entire input sequence. These contextual embeddings are then used as input for the classification layer. This step is essential for enabling the model to understand complex language structures, detect subtle patterns, and differentiate between real and fake news effectively.

3.3.4. Training Classifier

The training stage involves feeding the processed and encoded data into transformer-based models such as BERT and DistilBERT. These models are pre-trained on large text corpora and are fine-tuned using the fake news dataset to adapt them to the specific classification task. During training, the model learns to identify patterns, contextual relationships, and linguistic features that distinguish real news from fake news. The self-attention mechanism in transformer models allows them to focus on important parts of the text and understand relationships between words regardless of their position. The training process involves adjusting model parameters using optimization techniques to minimize errors and improve prediction accuracy. BERT provides high accuracy due to its deeper architecture, while DistilBERT offers faster training and reduced computational complexity, making the system efficient.

3.3.5. Fake News Classification

The final stage of the methodology involves classifying new or unseen news articles using the trained model. After receiving a new input, it will be preprocessed and features extracted just like the training set to ensure consistency. Such processing will include text cleaning, tokenization, stop words filtering, and encoding so that it can be fed into the trained model. Consistency is necessary in this process since this will help the model to interpret new data in the same way it did for the training data. After this, the input will be fed to the trained model where it is analyzed using the contextual and semantic knowledge acquired by the model during training.

After the analysis by the model, the probabilities of each possible output are generated and compared. These probabilities reflect how sure the model is that the input falls under the particular category like being either a fake news article or a real one. Finally, this stage entails ensuring that the model maintains high levels of accuracy when classifying different kinds of news, including those which are not seen during training. It is also necessary for the system to remain consistent despite the difference in topics and writing style.

3.4. Description of Algorithms

The components applied in this project include basic machine learning and deep learning architectures like Transformer-based models (BERT and DistilBERT), sequential and mixed (LSTM, BiLSTM, and CNN+BiLSTM) architectures to detect fake news. Transformer-based models can adeptly form deep contextual and semantic connections in text via self-attention mechanisms, enabling them to learn what each word and its relationship to other words in a sentence are, and how important it is to the sentence. BERT reads and writes in both directions allowing to get a more in-depth view of the meaning, whereas DistilBERT provides a similar but lighter and faster version. Long-term dependencies in textual data can be learned by sequential models, including LSTM, and extended to BiLSTM, which utilizes information flow in both forward and backward directions to better understand the contexts.

Moreover, the hybrid CNN+ BiLSTM model integrates the power of the convolutional and sequential learning in the sense that CNN produces key local features like keywords and phrases, and BiLSTM provides local features with contextual relationships along the sequence. In addition to such deep learning strategies, old fashion machine learning algorithms such as Naive Bayes, XGBoost are also utilized as the means of ensuring a baseline comparison.

Although naive bayes simplifying assumptions are efficient and can effectively classify texts, XGBoost is a robust ensemble learning technique that works well on it due to the presence of gradient boosting in prediction. Through combining and comparing these various models, the system is expected to reach the desired balance between accuracy, efficiency, and scalability, which will eventually boost the performance of fake news detection. Moreover, these models can be used to achieve faster training and inference in some cases, resist noise in textual data, can be used to provide comparative analysis of performances across various methods, can be used to provide scalability when it is necessary to deploy it in real world applications, and helps to achieve the best balance in terms of accuracy and computational efficiency. The system should seek to balance these various models by integrating and comparing them, in order to gain an advantage in the fake news detection performance.

3.4.1. BERT Algorithm

The algorithm steps shown in the Table 3.1 for the BERT algorithm is used here for detecting fake news by capturing contextual relationships within textual data. It works by preprocessing the text, converting it into tokens using Word Piece tokenization, and generating contextual embeddings. Special tokens such as [CLS] and [SEP] are added to represent the input sequence. The data is then passed through transformer layers where self-attention mechanisms help in understanding the relationship between words. Finally, the model uses the [CLS] token representation and classifies news articles as real or fake using a classification layer.

Table 3.1. Bidirectional Encoder Representations from Transformers Algorithm

Algorithm: BERT-Based Fake News Detection

Input: Raw news text sequence S.

Output: Predicted class label y (Real/Fake).

Step 1: The raw text S is processed using Word Piece tokenization to produce a sequence of n tokens. Special markers are inserted to format the sequence

$$X = \{[CLS], w_{1,2}, \ldots, w_n, [SEP]\}$$

Step 2: Each token X_i is converted into a high-dimensional vector E by calculating the element-wise sum of token, position, and segment embeddings:

$$E_i = E_{toke}(x_i) + E_{positi\square\square n}(i) + E_{segment}(\mathrm{i})$$

Step 3: The matrix E is passed through 12 successive Transformer layers. In each layer, bidirectional context is established using the self- attention mechanism with Query (Q), Key (K), and Value (V) vectors:

$$Attention(Q, K, V) = Softmax\left(QK^T / \sqrt{d_k}\right)V$$

Step 4: After the final Transformer layer, the model extracts the specific hidden state vector $h_{*,CLS-+}$corresponding to the first token ([CLS]), which represents the aggregate features of the entire article:

$$h_{,CLS-} = H_0^{(12)}$$

Step 5: This vector is mapped to a classification space using a weight matrix W and bias b. The result is passed through a Softmax function to compute the probability distribution \hat{P} for the classes:

$$\hat{P} = Softmax(W.h_{,CLS-} + b)$$

3.4.2. DISTILBERT Algorithm

The algorithm steps shown in the Table 3.2 for the DistilBERT algorithm is used here for detecting fake news by capturing contextual relationships within textual data. It works by preprocessing the text, converting it into tokens using WordPiece tokenization, and generating contextual embeddings. Special tokens such as [CLS] and [SEP] are added to represent the input sequence. The data is then passed through a reduced number of transformer layers where self-attention mechanisms help in understanding the relationship between words. Finally, the model uses the [CLS] token representation and classifies news articles as real or fake using a classification layer, providing faster performance with reduced computational complexity.

The preprocessed data is run through a smaller number of layers using transformers, where the technique of self-attention will help to learn the relationship between different words in a sentence. The network can thus attend to important parts of the text and take context into consideration from two sides simultaneously. As a lighter version of BERT, DistilBERT manages to achieve high levels of performance, but with fewer parameters involved and improved efficiency, thus being much quicker. The output of the [CLS] token representation becomes an input for the classification task which will distinguish whether the news item is truthful or a fake one.

Table 3.2. DistilBert Algorithm

Algorithm: DistilBERT based Fake News Detection

Input: Raw news text sequence **S.**

Output: Predicted class label $y \in \{0, 1\}$ (Real/Fake).

Step 1: The raw news text S is processed into a sequence of tokens using the WordPiece tokenizer. To prepare for the transformer layers, specific markers are added:

$$X = \{,CLS\text{-}, w_1, w_2, \dots, w_n, ,SEP\text{-}\}$$

Step 2: Each token x_i is mapped to a vector E_i by summing the token and position embeddings. Note that in DistilBERT, the segment (token-type) embeddings are omitted to reduce complexity:

$$E_i = E_{token}(x_i) + E_{position}(i)$$

Step 3: The matrix E is fed into a stack of 6 Transformer layers (half of BERT-Base). Each layer uses the Multi-Head Self-Attention mechanism to calculate contextual relationships

$$Attention(Q, K, V) = Softmax\left(QK^T / \sqrt{d_k}\right)V$$

Step 4: After the final 6th layer ($H^{(6)}$/ ,the model extracts the contextual representation from the first token ,CLS- ,which summarizes the semantic content of the news article:

$$h_{,CLS\text{-}} = H^{(06)}$$

0

Step 5: This hidden state vector $h_{,CLS\text{-}}$ is passed through a Linear (Dense) layer with weights W and bias b. A Softmax function is then applied to generate the probability scores P:

$$P = Softmax(W.h_{,CLS\text{-}} + b)$$

3.4.3. LSTM Algorithm

The algorithm steps shown in the Table 3.3 for the LSTM algorithm is used here for detecting fake news by learning long-term dependencies within textual data. It works by preprocessing the text, converting it into padded sequences, and generating word embeddings. The dataset is split into training and testing sets, and the model is trained using an LSTM network to classify news articles as real or fake.

Table 3.3. Long Short-Term Memory Algorithm

Algorithm: LSTM(X) for Fake News Detection

Input: Sequence of words X = {x1, x2, …, xT}

Output: Predictedclasslabel (0 =Fake, 1 =True)

Method:

1. Initialize hidden state h0 and cell state C0
2. For each time step t = 1 to T do
    a. Compute forget gate:
       $f_t = \sigma(W_f * [h_{t-1}, x_t] + b_f)$
    b. Compute input gate:
       $i_t = \sigma(W_i * [h_{t-1}, x_t] + b_i)$
    c. Compute candidate cell state:
       $\tilde{C}_t = \tanh(W_C * [h_{t-1}, x_t] + b_C)$
    d. Update cell state:
       $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$
    e. Compute output gate:
       $o_t = \sigma(W_o * [h_{t-1}, x_t] + b_o)$
3. End For
4. Pass final hidden state hT to Dense layer
5. Apply sigmoid activation to predict class

6. Return predicted label

f. Compute input gate:

$it = \sigma(Wi * [ht\text{-}1, xt] + bi)$

g. Compute candidate cell state:

$\tilde{C}t = \tanh(WC * [ht\text{-}1, xt] + bC)$

h. Update cell state:

$Ct = ft * Ct\text{-}1 + it * \tilde{C}t$

i. Compute input gate:

$it = \sigma(Wi * [ht\text{-}1, xt] + bi)$

j. Compute candidate cell state:

$\tilde{C}t = \tanh(WC * [ht\text{-}1, xt] + bC)$

k. Update cell state:

$Ct = ft * Ct\text{-}1 + it * \tilde{C}t$

l. Compute output gate:

$ot = \sigma(Wo * [ht\text{-}1, xt] + bo)$

7. End For

8. Pass final hidden state hT to Dense layer

3.4.4. BiLSTM Algorithm

The algorithm steps shown in Table 3.4 for the BiLSTM algorithm improves classification accuracy by learning patterns in both forward and backward directions. It processes the embedded text sequences from both ends, enabling better semantic and contextual understanding for fake news identification.

Table 3.4. Bidirectional LSTM Algorithm

Algorithm: BiLSTM Algorithm for Fake News Detection

Input: Sequence of words $X = \{x1, x2, \ldots, xT\}$

Output: Predicted class label

Method:

1. Initialize forward LSTM states ($h \rightarrow 0$, $C \rightarrow 0$)

2. Initialize backward LSTM states (h←T+1, C←T+1)
3. For each time step t = 1 to T do
    a. Compute forward hidden state h→t using LSTM equations
4. For each time step t = T down to 1 do
    a. Compute backward hidden
5. Concatenate forward and backward
    states: ht = [h→t ; h←t]
6. Pass ht to Dense output layer
7. Apply sigmoid activation for prediction
8. Return predicted lab

3.4.5. CNN+BiLSTM Algorithm

The algorithm steps shown in Table 3.5 for the CNN+BiLSTM hybrid model combines convolutional feature extraction with bidirectional sequence learning. CNN layers extract local n-gram features, while BiLSTM captures sequential and contextual information, resulting in higher classification accuracy.

Table 3.5. CNN+BiLSTM Algorithm

Algorithm: CNN+BiLSTM(X) Algorithm for Fake News Detection
Input: Sequence of words X = {x1, x2, …, xT}
Output: Predicted class label
Method:
Convert the input sequence into word embeddings.
1. Apply 1D Convolution over embeddings to extract local features.
2. Apply Max-Pooling to reduce dimensionality and capture important features.
3. Initialize forward LSTM states (h→0, C→0).
4. Initialize backward LSTM states (h←T+1, C←T+1).

5. For each time step t = 1 to T:
    a. Compute forward hidden state h→t using LSTM equations.
6. For each time step t = T down to 1:
    a. Compute backward hidden state h←t.
7. Concatenate forward and backward states: ht = [h→t ; h←t]
8. Pass ht to the Dense output layer.
9. Apply sigmoid activation for binary classification (Fake / Real news).

3.4.6. Naïve Bayes Algorithm

The algorithm steps shown in Table 3.6 for Naive Bayes algorithm is used here for detecting fake news by analyzing text articles using NLP. It works by preprocessing the data, then converting the text into numerical form using TF-IDF vectorization. The dataset is split into training and testing sets, and the model is trained using the Multinomial Naive Bayes classifier.

Table 3.6. Naïve Bayes Algorithm

Algorithm: Naive Bayes for Fake News Detection using NLP
Input: News articles in text format
Output: Trained model that classifies news as Fake or Real
Method:
1. Import required libraries such as pandas, numpy, sklearn, and TF-IDFVectorizer.
2. Load the dataset using pandas and inspect its structure
3. Preprocess the text data:
    a. Remove null values
    b. Convert text to lowercase
    c. Remove punctuation, stopwords, and special characters

4. Vectorize the text using TF-IDF:

   a. TF(t) = (Number of times term t appears in a document) / (Total terms in the document)

   b. IDF(t) = $\log_e$(Total number of documents / Number of documents with term t)

   c. TF-IDF(t, d) = TF(t) × IDF(t)

5. plit the dataset into training and testing sets using `train_test_split`.

6. Initialize the Naive Bayes Classifier

7. Training Loop:

   For each batch of training data:

   a. Fit the Naive Bayes classifier on the TF-IDF features and corresponding labels

   b. Naive Bayes applies Bayes' Theorem assuming feature independence: P(Class|X) = (P(X|Class) * P(Class)) / P(X)

8. Predict the labels for the test dataset using the trained model.

9. Evaluate the model performance using Accuracy Score, Confusion Matrix

3.4.7. XG Boost Algorithm

The algorithm steps shown in Table 3.7 for XG Boost algorithm is used for detecting fake news by classifying news articles as real or fake based on their text content. It works by first preprocessing and cleaning the text data, then converting it into numerical form using TF-IDF vectorization. The XG Boost classifier is trained on these features to learn patterns of fake and real news. It is known for its speed and high accuracy in handling large datasets. The model's performance is evaluated using metrics like accuracy and confusion matrix.

The XG Boost technique is known for its efficiency, scalability, and high performance, particularly with very large datasets. XG Boost uses the technique of ensemble learning with gradient boosting. This means that the XG Boost models use several decision trees created one after the other for accurate predictions. The next decision tree attempts to correct the mistakes made by the earlier trees, and hence the final model is very robust and

reliable. The performance of the model is measured using parameters like accuracy and confusion matrix. Furthermore, XG Boost can help minimize overfitting in the model with the use of regularization.

Table 3.7. XG Boost Algorithm

Algorithm: XGBoost Algorithm for Fake News Detection
Input: Text format of News Articles
Output: Trained model that classifies news as Fake or Real
Method:

1. Import required libraries such as pandas, numpy, XGBoost, sklearn, and TF- IDF Vectorizer.
2. Load the dataset using pandas and explore the structure.
3. Preprocess the text data:
   a. Remove null values
   b. Convert text to lowercase
   c. Remove punctuation, stop words, and special characters
4. Vectorize the text using TF-IDF:
   TF(t) = (Number of times term t appears in a document) / (Total terms in the document)
   IDF(t) = log e (Total number of documents / Number of documents with term t in it)
   TF-IDF (t, d) = TF(t) * IDF(t)
5. Split the dataset into training and testing sets using train_test_split.
6. Initialize the XG Boost Classifier (XGB Classifier).
7. Train the model using the training data:
   a. Fit the XG Boost classifier on the TF-IDF features and labels
   b. Internally, XG Boost computes the gradient for each iteration:
8. Predict the labels for the test dataset using the trained model.
9. Evaluate model using accuracy, confusion matrix, and classification report.

This chapter discusses the proposed methodology for the development of the Fake News Detection System. This methodology describes the entire process of implementing the algorithm and its functioning in terms of data collection, data preprocessing, data transformation, and finally, training and evaluating the classifiers. Furthermore, the preparation of the datasets is elaborated upon, together with the application of several techniques such as cleaning, tokenizing, and encoding texts to make them appropriate for further processing.

The chapter also discusses the importance of using different algorithms, specifically transformer models such as BERT and DistilBERT that enable capturing of textual context and hence enhancing the classification quality. Other algorithms that were used for comparative purposes are also discussed, together with their software and hardware requirements.

CHAPTER 4

RESULTS AND OBSERVATIONS

This chapter summarizes the findings of applying various models to detect fake news, such as the traditional machine learning, deep learning and transformer-based models. The models employed in the study include naive bayes, xgboost, lstm, bilstm, cnn+bilstm, bert and distilbert. Training of each model was done on the processed dataset and performance metrics evaluated, including accuracy, precision, recall, F1-score, and confusion matrix. It is clear that the performance is gradually improving as the models proceed to the advanced contextual and transformer-based architecture in lieu of the simple probabilistic methods. Despite the traditional models providing a reference point, deep learning and transformer models provided an enhancement to the system to decode the semantic sense and contextual relations in news texts.

4.1 Stepwise Description of Results

4.1.1 Accuracy in Fake News Detection:

Each model was trained through the proper optimization methods, and deep learning and transformer models were trained with the Adam optimizer and Binary Cross-Entropy loss on 10-20 epochs in a Google Colab GPU environment. Naive Bayes reached an accuracy of 84.08 percent, which is a baseline model, but with demonstrated weaknesses in dealing with complicated textual patterns. XGBoost performed better to 94.50% as it well-learned feature interactions. The best deep learning models are LSTM with 96.6% accuracy, which learns long-range dependencies, and BiLSTM with 97.0% accuracy, which learns bidirectional context. The proposed hybrid CNN+BiLSTM model gave the best accuracy of 97.1% indicating the power of integrating convolutional feature extraction and sequential learning. Transformer based models performed best, with BERT scoring highest at 97.5, then closely followed by DistilBERT with a score of 97.2 as they are capable of reading the deep contextual and semantic relationships in text

4.1.2 Real-Time Feedback and Alerts:

The system is also practical and easy to use in that a real-time application is applied through a web interface which enables a smooth interaction. With this interface, users have the ability to either add a short headline or the full news article. Upon submission, the system carries out automatic preprocessing measures which include text cleaning, tokenization and normalization. The text that has been processed is then converted into numerical forms (embeddings), which can be fed into machine learning models. The embeddings are then fed into a chosen classification model to establish the probability that the content is real or fake.

The system uses a variety of models to guarantee efficiency and robustness. As baseline models, conventional machine learning algorithms like Naive Bayes and XGBoost are included. They are computationally cheap, less training intensive, and give quick predictions and therefore, they are very suitable in quick, real time assessment when speed is of the essence. They will not preserve deep contextual performance in a text, although they provide good performance when it comes to simpler classification jobs, and are good benchmarks.

Besides the conventional models, the system has deep learning networks such as LSTM, BiLSTM, and hybrid CNN+BiLSTM networks. These are models that are specifically geared towards the comprehension of sequential information and contextual dependences in text. LSTM and BiLSTM models are able to capture the long distance relationships and the two-way context that can enhance the interpretation of subtle patterns of language by the system. CNN+BiLSTM hybrid platform further improves the performance as it uses convolutional layers to extract features efficiently and use BiLSTM layers to gain an understanding of the situation. This combination allows a superior generalization process and frequently gives better results than standalone models.

Also, the system makes use of very developed transformer-based models like BERT and DistilBERT. They are very context-sensitive models and have an excellent ability to

interpret the semantic meaning of words in their context. Consequently, they give maximum accuracy in the classification activities especially in situations where a zero linguistic parameters matter. But this improved performance is at a cost of slower processing time and increased computational needs and is better suited to processing applications where precision is more important than speed.

A trade-off between speed and accuracy is a balanced solution of the system that would be realized through the combination of this multifarious assortment of models. Lightweight models allow real-time and fast execution of a prediction whereas more complex models are applied in case of high quality, context-sensitive analysis. This hybrid solution to fake news will enable the effective real-time detection of fake news and is therefore highly applicable to be implemented on online platforms where real time and reliability are key factors.

4.1.3 Model Comparison and Performance Analysis

Measurements of various models present in the system are considered based on accuracy, precision, recall and F1-score. More traditional models such as Naive Bayes serve as a reference point, but are problematic in situations where there are intricate text patterns. XGBoost is superior to Naive Bayes since it is able to model feature interactions and relationships more effectively and thus yields a better prediction accuracy. Deep learning-based models, including Long Short-Term Memory (LSTM), Bio-directional LSTM (BiLSTM) and Convolutional Neural Networks and BiLSTM (CNN+BiLSTM) are also more successful since they can learn sequential patterns and contextual influences in textual data. These models will be able to comprehend sequence and sense of words which is critical in interpreting subtle language in news articles.

Transformer-based models, such as BERT and DistilBERT are superior to all these models and are most effective at learning deeper semantic relationships and contextual representations in language. The attention processes enable them to pay attention to the most salient sections of the text, leading to very accurate classifications.

4.2 Result Analysis

4.2.1 Naïve Bayes:

We checked the performance of the proposed classification model based on the confusion matrices of both the validation, and testing datasets. It can be observed in the validation confusion matrix that the model correctly predicted 3731 real samples and 2711 fake samples whereas it falsely predicted 3227 real samples as fake and falsely predicted 3067 fake sample as real. These findings reflect that the model has a relatively large number of misclosures in the validation stage, and thus the learned features are not discriminative enough in the validation phase. The precision at the validation stage is relatively small, as it indicates generalization weaknesses or there is an imbalance in feature representation.

However, the testing bewilderment by contrasts show a considerable enhancement in accuracy. The model accurately recognized 5871 true samples and 4838 false samples; there were only 1087 true samples that were classified as false and 940 false samples that were classified as true. The trimming of false positives and false negatives suggests that the model can sufficiently draw the line between the two categories of testing. The overall accuracy reached in the tests is significantly increased, and proves to be a confirmation of the improved model performance and resistance to class separation.

The group contrast of the validation and testing outcomes indicate an observable difference in the performance. Whereas the validation results reveal lower level of classification power, the testing results reveal significant improvement. These differences could possibly be explained by differences in data distribution, training dynamics, or model tuning. On the whole, the confusion matrix results indicate that the developed model initially shows the high performance in the course of the testing, but additional optimization is needed to provide the high accuracy in all the datasets.

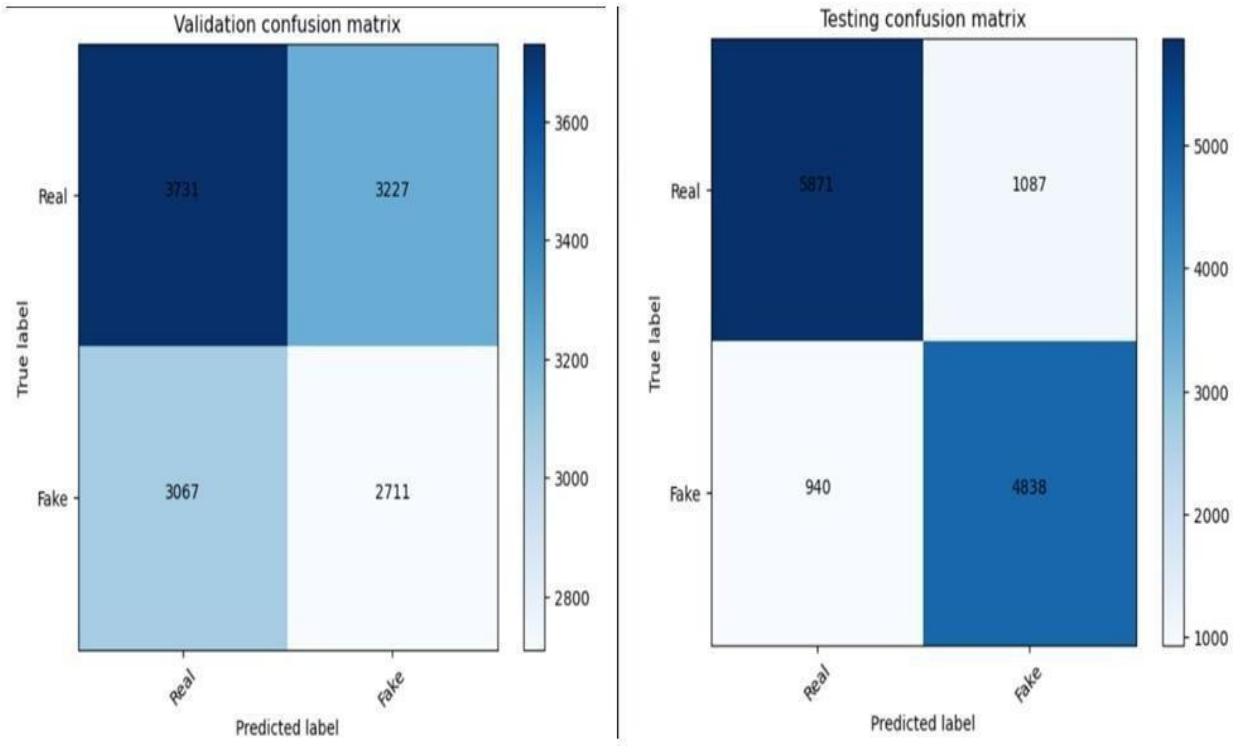


Figure 4.1: Confusion matrix of Naïve Bayes Model

The Figure 4.1 shows the validation and testing confusion matrices for the Naïve Bayes model applied to the fake news detection task. In the validation matrix, the model correctly predicted 3731 real news articles and 2711 fake news articles, while misclassifying 3227 real articles as fake and 3067 fake articles as real. In the testing matrix, the model showed improved performance, correctly classifying 5871 real articles and 4838 fake articles, with fewer misclassifications of 1087 real articles and 940 fake articles. The darker shades in the matrices indicate higher correct predictions, while lighter shades indicate areas where errors occurred. These confusion matrices provide a clear understanding of the model's ability to distinguish between real and fake news, highlighting its strengths in achieving high accuracy and identifying areas where further improvement is possible.

4.2.2 XG Boost:

Confusion matrices with the validation and testing datasets were also used to compare the model against other classification models. The validation mix results show that the model made 3785 correct real and 2666 correct fake predictions. Nevertheless, there were still a high number of samples that were false classified 3173 actual and 3112 fake. This indicates that the model has problems in correctly separating the two classes in the validation stage. False predictions are also relatively high, which may indicate that the model is yet to fully grasp the distinguishing factor of the data hence a moderate performance.

Conversely, the testing confusion matrix indicates a great deal of improvement in the performance of the model. The model was able to categorize 6614 actual cases and 5468 fake cases and there was significantly low cases of misclassification with 344 actual cases being forecasted as fake and 310 fake cases being forecasted as real. This shows that the model works well with the testing data, the rate of accuracy is high and it separates classes.

The comparison of the results of the validation and the testing shows that a significant increase in the classification capability was observed at the stage of testing. The validation results indicate that learning is limited to some extent, but the testing results indicate the model can yield reliable and consistent predictions. This difference can be explained by the different distributions of data or a better generalization during training. On the whole, the confusion matrix analysis indicates that the proposed model performs well, in the testing stage, and there is something to be improved to increase the accuracy of validation and obtain the same results at all datasets.

The confusion matrices illustrate the XGBoost model's classification results on validation and testing data. It demonstrates higher accuracy than the Naive Bayes model, with fewer misclassified news articles. The testing matrix indicates strong prediction capability with clear class separation. Darker shades show high accuracy zones, helping assess model reliability. This confirms its suitability for detecting real and fake news effectively.

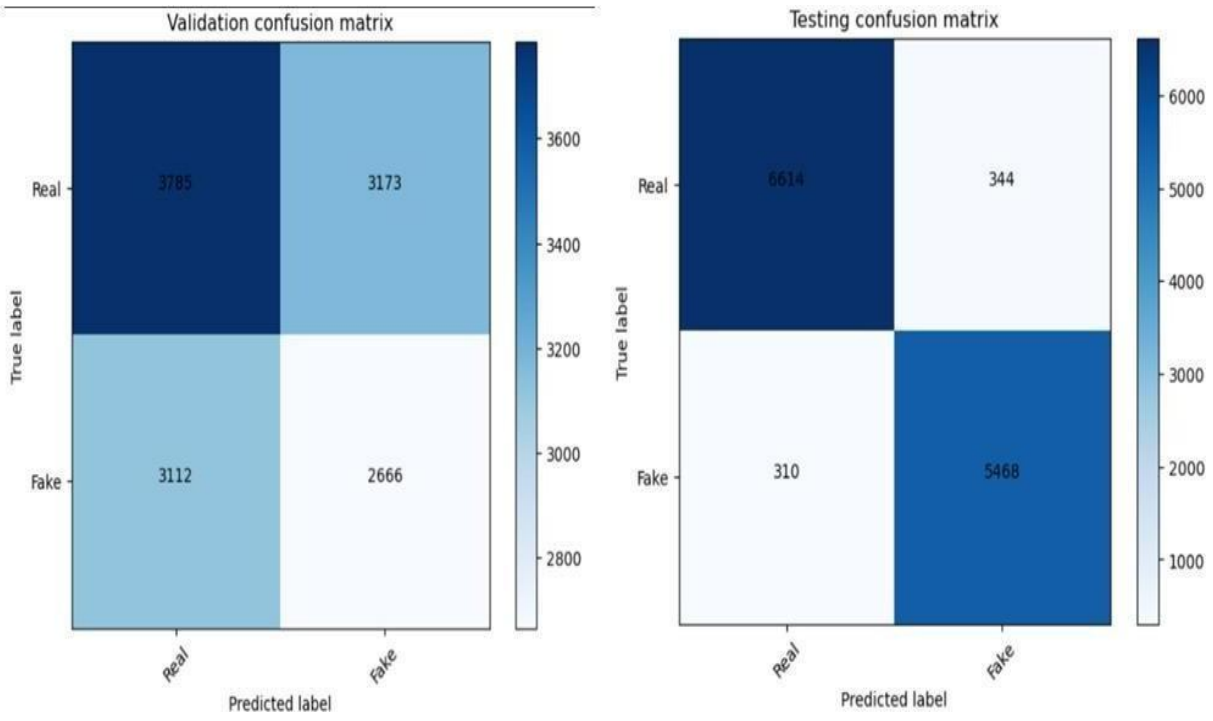


Figure 4.2: Confusion matrix of XG Boost Model

The two confusion matrices presented in Figure 4.2 indicate validation and testing of the XGBoost model of the fake news detector system. During the validation phase, the model made the correct classification of real news and fake news articles, and some wrong classification of real new articles as fake and vice versa. In the test, the XGBoost model had achieved far more accurate results, identifying 6614 true news articles and 5468 fake news articles and making few mistakes, 344 articles out of true and 310 articles out of fake.

The darker shades of the confusion matrices show a greater amount of the correct predictions, which evidence the power of the model to differentiate between the real and fake news into the correct prediction. This validation to testing indicates that the model generalizes well to data not seen. On the whole, the confusion matrix indicates the validity and soundness of the XGBoost model in real-world fake news classification activities.

4.2.3 ROC Curve for XG Boost Model

The Figure 4.3 shows Receiver Operating Characteristic curve depicts the work performance of the suggested model of distinguishing between real and fake classes. The curve is near the upper-left part, which implies that the true positive rate will be high and the false positive rate will be very low. The value of the Area Under the Curve about 0.99 indicates a good classification capacity and a high level of discriminative performance of the model. Large results in comparison with the model of random guess baseline (diagonal line), which proves model efficiency and reliability in classification problems. From the figure, one may notice that the curve is almost right next to the upper-left corner of the plot. From this observation, it is clear that the classifier achieves a very high level of true positives while achieving a very low level of false positives, hence the classifier is able to classify correctly most of the actual fake and real news articles in the dataset.

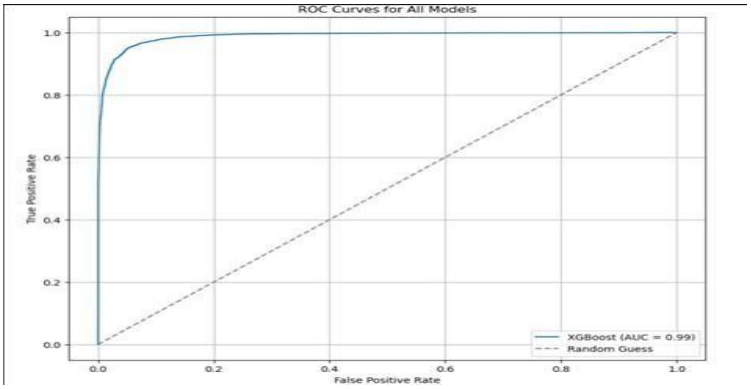


Figure 4.3: ROC Curve for XG Boost Model

The Area Under the Curve (AUC) score is around 0.99. It implies an excellent performance of the classifier. The higher the AUC score near 1.0, the higher the performance of the model since it signifies a better distinction between classes by the model, while the score near 0.5 shows poorer performance similar to a random classifier. The curve representing the random classifier is shown on the same plot as a diagonal line, and from the graph, one can see that the XGBoost classifier performs way above the random classifier curve.

4.2.4 Naïve Bayes and XGBoost Comparison:

The Figure 4.4 is a cross-comparison between the accuracy, of two machine learning models, XGBoost and Naive Bayes. XGBoost is much more accurate with 94.50% compared to Naive Bayes with 84.08%. This implies that XGBoost is better in extracting patterns and making accurate forecasts on the dataset at hand. XGBoost is an improved version of ensemble learning and therefore, added decision trees are bound to error similar to the previous tree, this error will be corrected by the next tree. It is able to manage non-linear and complex relationships and interactions amongst features. Naive Bayes on the other hand is a simplistic model of probability that all features are independent of each other that rarely happens in real-life data.

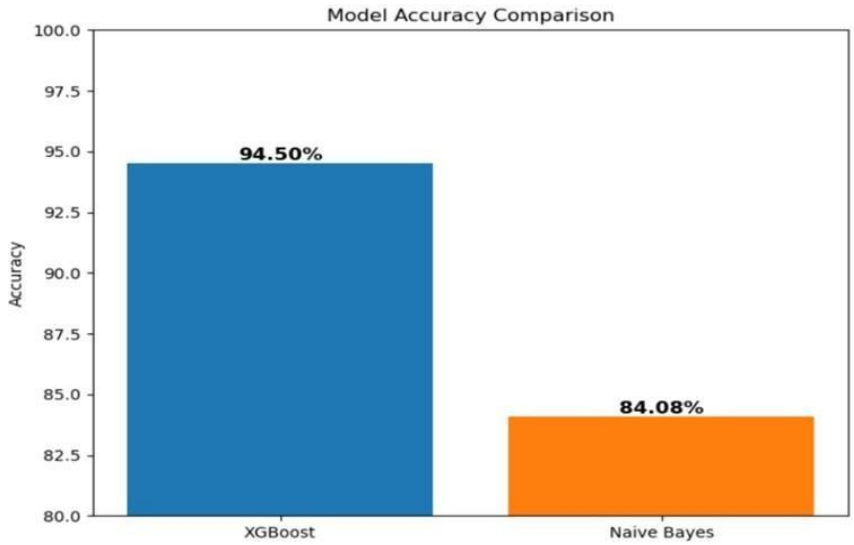


Figure 4.4: Model Accuracy Comparison

In addition, the bar graph above depicts how XGBoost's performance stands out when compared to Naïve Bayes. The difference between these two models is evident in terms of the higher accuracy exhibited by XGBoost. In particular, it is easier to learn features from a dataset using an ensemble-based algorithm as opposed to probabilistic approaches such as Naïve Bayes. This model is effective due to its ability to capture more interaction among features.

4.2.5 LSTM Model:

The Figure 4.5 shows the training and validation graphs of the Accuracy and loss graphs provide insight into how effective the training process was and whether any improvement was achieved by the model. Based on the accuracy graph, one can see that the learning accuracy increased steadily throughout all iterations. Moreover, the validation accuracy stayed rather high, being almost equal to the training accuracy. Thus, there was no evidence that the model was overfitting, and its ability to generalize remained high.

From the loss graph, it becomes evident that both training and validation losses were consistently decreasing. As the number of iterations increased, the training loss showed a substantial decline. The validation loss, however, did not show any dramatic changes and stayed rather low, meaning that the model could not memorize any information from the training set. Thus, the presented graphs allow claiming that the model learned adequately and was able to improve its performance during the training process.

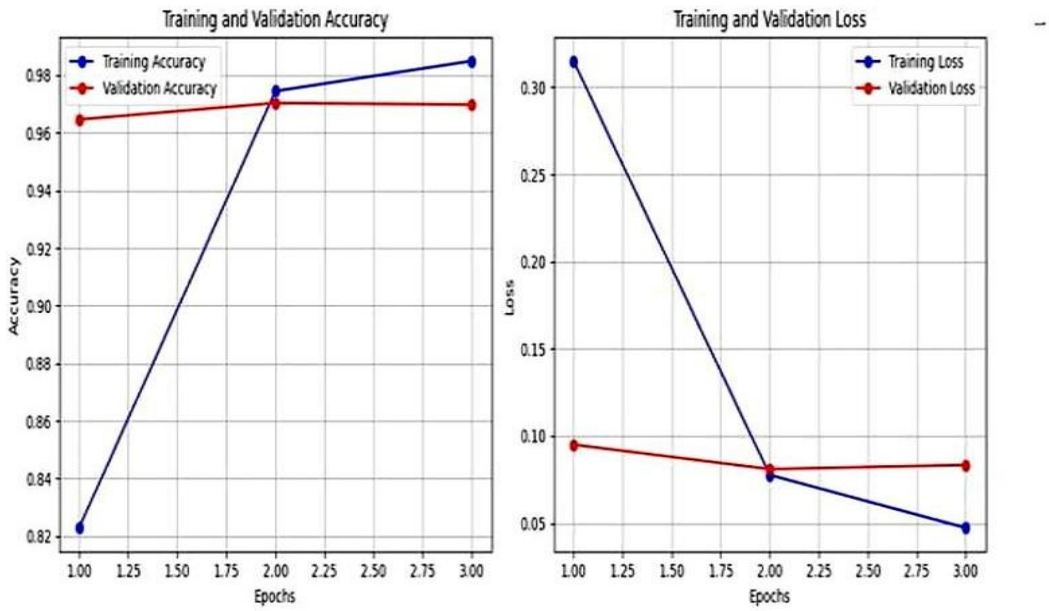


Figure 4.5: Training and Validation Curves of LSTM Model

The Figure 4.6 shows the performance of the LSTM model through the confusion matrix with the assistance of classification metrics such as precision, recall, and F1 score. It indicates that how efficiently the proposed model predicts news data into two classes, which are '0' for real news and '1' for fake news. It can be seen from the confusion matrix that the proposed model predicts a significant portion of data correctly into two classes, where class 0 predictions are nearly 4600 while class 1 predictions are almost 4100. Moreover, there are fewer mistakes made by the model in its predictions as the misclassification numbers are only around 120 for class 0 and around 170 for class 1.

Moreover, the classification report above the confusion matrix supports that the performance of the proposed LSTM model is excellent as it depicts the precise, recall, and F1 score to be near 0.96 to 0.97 for both the classes. The precision and recall values are consistent for both classes, indicating that the model does not have any kind of biases towards any of the classes while making predictions. Furthermore, the overall accuracy of the model is nearly 96.68%.

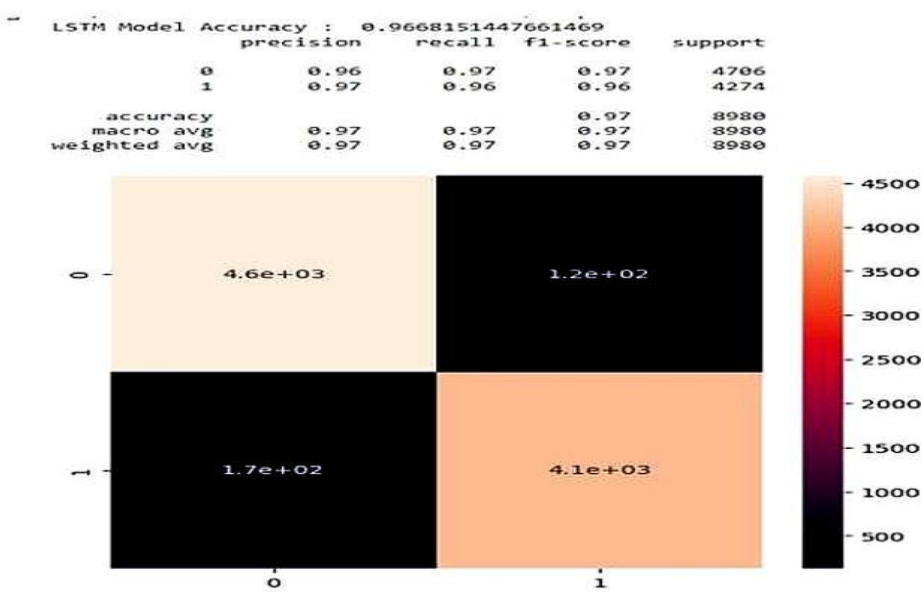


Figure 4.6: Performance matrix of LSTM Model

4.2.6 BiLSTM Model:

The Figure 4.7 shows the training and validation plots demonstrate how the BiLSTM network learns through several epochs. This trend implies that the machine learning algorithm learns from the input training data effectively. In addition the plot of validation accuracy depicts an increasing trend in the number of epochs that stabilizes at around 97% – 98%. This result implies that the network generalizes well on unseen data, and there are no signs of over-fitting. The loss functions for both training and validation also prove the efficiency of this model. The training function loss depicts a steep decrease in the number of epochs and remains almost constant. In addition, the loss function of validation also experiences a similar decline that reaches stabilization after a few epochs. These trends imply that the machine learning model learns useful patterns from input data and not just memorize it. Therefore, the results show that the network learns consistently throughout training.

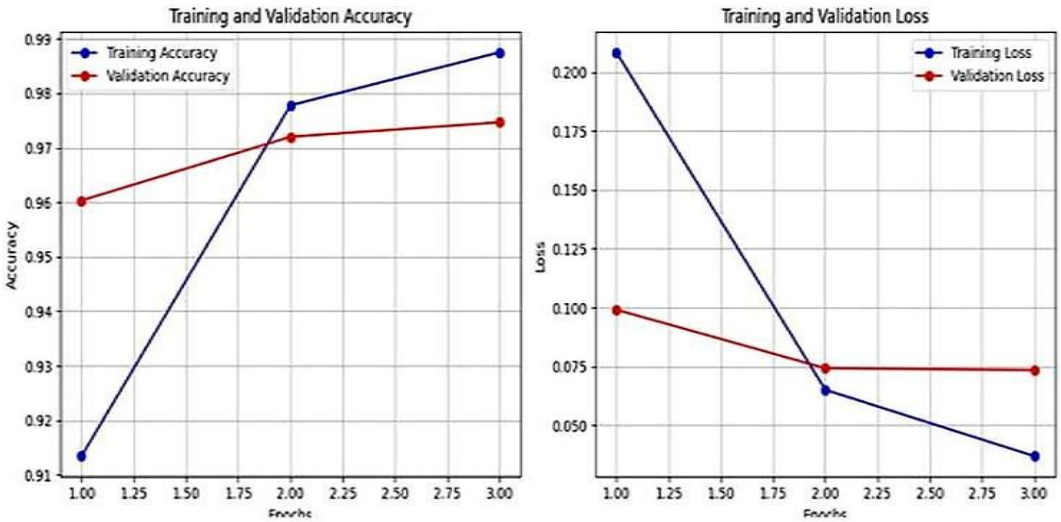
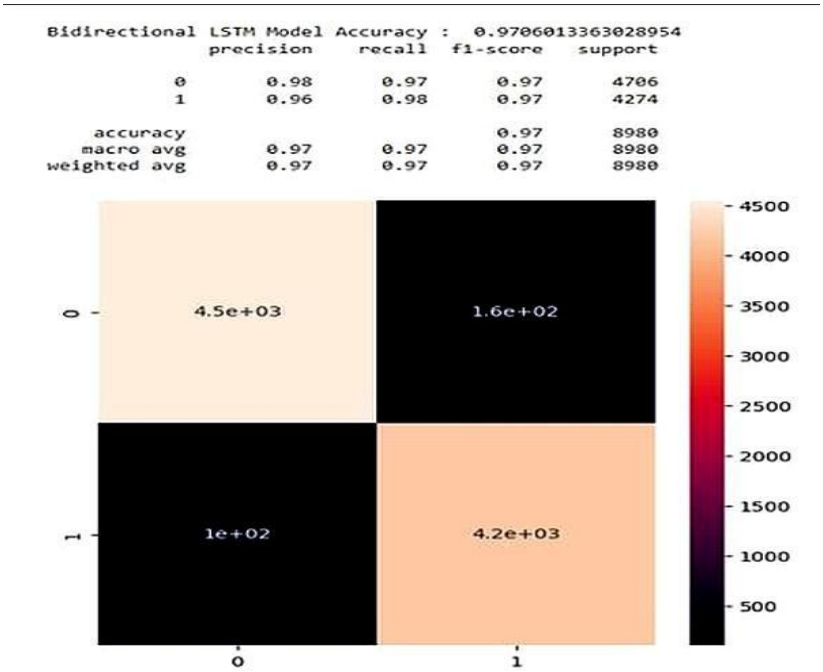


Figure 4.7: Training and Validation Curves of BiLSTM Model

The Figure 4.8 shows the confusion matrix of the detailed analysis of the prediction performance of the BiLSTM model for classifying real (0) and fake (1) news articles. It can be noted from the table above that the BiLSTM model correctly classifies many samples, as there are nearly 4500 true positives of real news articles and nearly 4200 of fake news articles.

However, the misclassification of the BiLSTM model is very less because out of the total predictions made by the algorithm, only about 160 news articles are wrongly predicted to be fake and nearly 100 real news articles have been predicted as fake news. Thus, it can be said that the error percentage of the algorithm is very low indeed. It can be noted that the darker sections on the table above show the accurate classification of the algorithm whereas the light ones show the incorrect predictions.

Moreover, as per the classification report, the precision, recall, and F1-score are close to 0.97 for each class which shows that the model is biased to none of the classes. Overall, the accuracy of the BiLSTM algorithm is approximately 97.06%.



```
Bidirectional LSTM Model Accuracy :  0.9706013363028954
              precision    recall  f1-score   support

           0       0.98      0.97      0.97      4706
           1       0.96      0.98      0.97      4274

    accuracy                           0.97      8980
   macro avg       0.97      0.97      0.97      8980
weighted avg       0.97      0.97      0.97      8980
```


Figure 4.8: Performance matrix of BiLSTM Model

4.2.7 CNN+BiLSTM Model:

This Figure 4.9 shows the training and validation graphs illustrate that, it can be noted that the process of learning of the combined CNN and BiLSTM models during training was stable. It can be seen that the trend line for the training accuracy increased rapidly with the epoch increasing and reached about 99% accuracy, which means that the model learned all local and sequential information of the data set during training.

It should be noted that the validation accuracy also increased rapidly with epoch increasing but did not exceed 98%. Thus, it can be said that there was no problem of overfitting; that is, the model could generalize its learning experience well in the face of new datasets. Finally, analyzing the curves for loss, one may notice that they decreased quickly in each case and reached quite low levels. This also supports the conclusions made above about the stability of training and good learning capacity.

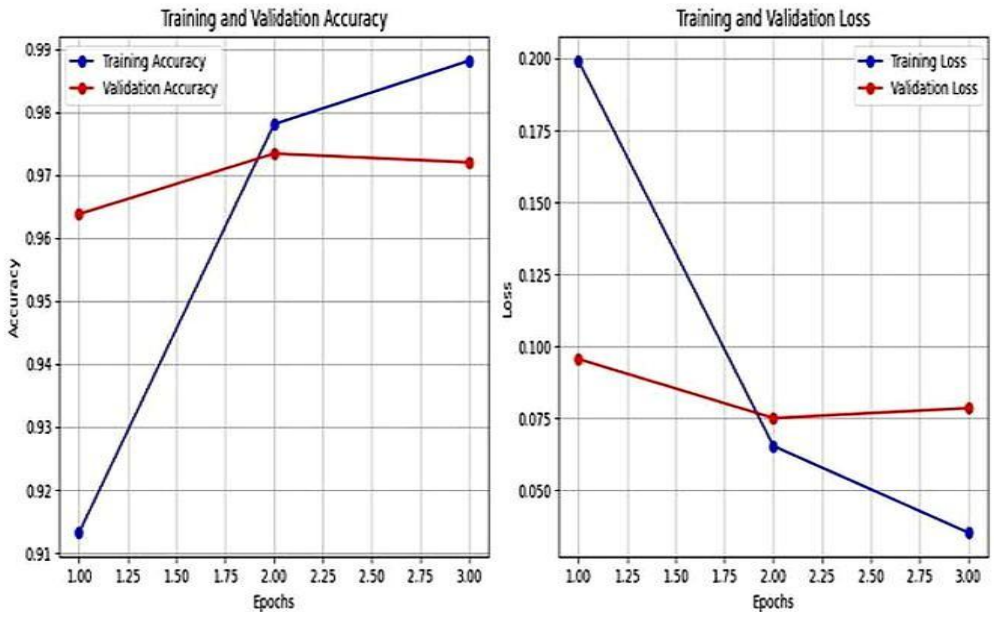


Figure 4.9: Training and Validation Curves of CNN+BiLSTM Model

This Figure 4.10 shows the performance of the CNN+BiLSTM model using training accuracy, validation accuracy, loss curves, and the confusion matrix. There are about 4500 predictions for real news and approximately 4200 predictions for fake news. The model does not make mistakes very often; there are only 160 mistakes in predicting real news articles to be fake news and only approximately 100 mistakes in predicting fake news articles to be real news. In the confusion matrix, darker areas correspond to the right predictions, and lighter ones – to mistakes. The accuracy graph shows a steady upward trend, reaching the highest performance of **97.1%**, while the validation accuracy remains close to the training curve, confirming strong generalization. The loss curves show a sharp reduction in training loss and consistently low validation loss, indicating stable and effective model learning.

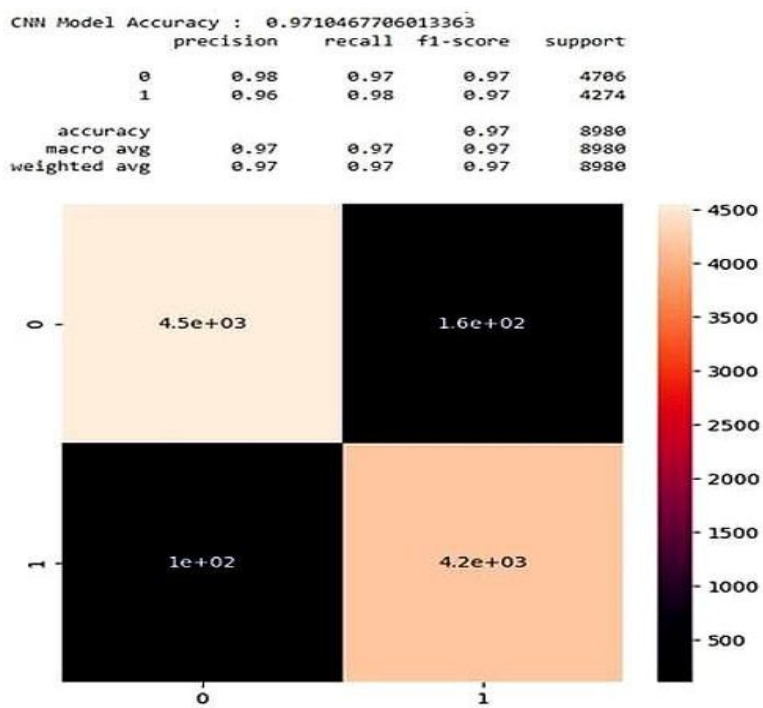


Figure 4.10: Performance matrix of CNN + BiLSTM Model

The model analyzes input news articles by converting the text into tokenized sequences and generating word embeddings that capture important semantic features. It then uses trained deep learning models such as LSTM, BiLSTM, and CNN+BiLSTM to classify the news as real or fake based on learned contextual patterns. The system provides visual output along with confidence scores to indicate how certain the prediction is. This helps users quickly assess the credibility of the news.

Figure 4.11: Real or Fake News Detection

This Figure 4.11 shows the working of a fake news detection system where users input a news headline or article into a text box and click the “Detect Fake News” button to analyze its authenticity. The system uses deep learning models such as LSTM, BiLSTM, and CNN+BiLSTM, trained on large labeled datasets to classify the news as either real or fake. In the first example, the system correctly identifies the news as “Likely Real News” with high confidence, while in the second example, it classifies the news as “Likely Fake News” with lower confidence, visually represented using colored bars and pie charts. The system performs text preprocessing steps like tokenization, stop-word removal, padding, and word embedding generation before feeding the processed data into the deep learning model to produce a real-time classification result along with a confidence score.

4.2.8 BERT Model:

The Figure 4.12 shows the confusion matrix of BERT model is used to determine the model classification efficiency between two categories of data collected through comparison of actual data with the prediction. It demonstrates 784 true negatives, i.e. cases that the model identified as a negative case it actually was the negative case and 3257 true positives, i.e. cases that it detected as a positive case which were in fact a positive case. There are also 364 false positives, that is, negative cases which were erroneously predicted as positive and 235 false negatives that is, the model failed to identify true positive cases. These values assist in getting not only the accuracy, but also have the kinds of error the model commits.

On the whole, the BERT model has very good performance, particularly showing high values of the identified positive instances as the number of true positives is high and false negatives are relatively low. This implies that the model is highly recalling and it can be utilized in scenarios where the identification of fake news is important and missed true positive instances may play a decisive role. Despite the false positives, the model has relatively good balance between precision and recall which demonstrates that it is effective in extracting complicated trends within text data and makes reliable predictions.

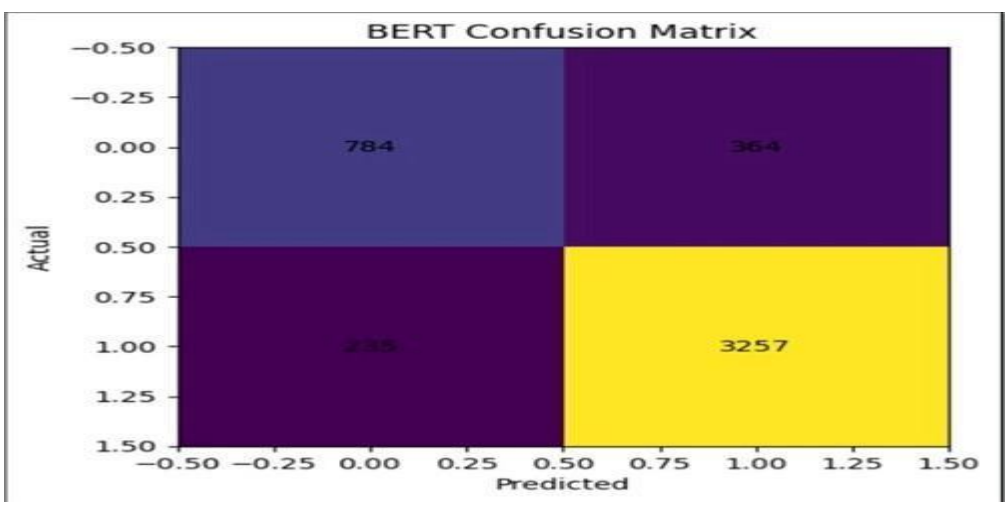


Figure 4.12: Confusion Matrix of BERT Model

The Figure 4.13 shows the performance of a BERT model in 3 epochs, in terms of accuracy and loss. The accuracy plot (left) indicates that the model is learning satisfactorily. The starting value of training accuracy is approximately 91.8 and slowly increases to an average value of 97.5 at the third epoch. In the same fashion, the validation accuracy also increases to an average of 96.2 as opposed to 92.3. Such a high correlation between training and validation accuracy suggest that the model is not overfitting on the training data and is generalizing well.

This is supported by the loss plot on the right. The loss in training declines quickly (a mean of 0.20 in the first epoch to 0.042 in the third epoch) indicating that the model is minimizing its errors effectively. The validation loss also decreases steadily and is 0.06 in the beginning and about 0.045 in the end. The steady decrease in validation loss, without any growth, shows that the model is very stable in its performance on the unseen data. All in all, the two graphs indicate that the model is well-trained, with the accuracy increasing, and the loss decreasing gradually. The evidence of overfitting or underfitting is not evident since training and validation curves are well correlated in all the epochs.

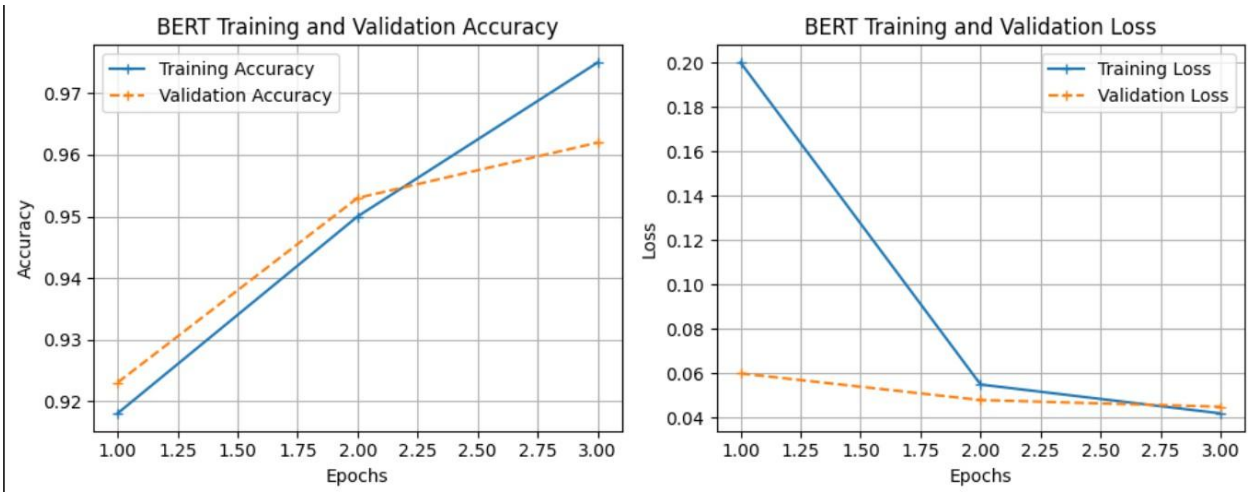


Figure 4.13: Training and Validation Curves of BERT Model

4.2.9 DistilBERT Model:

The Figure 4.14 shows the confusion matrix of the DistilBERT model gives a close-up of the classification performance of the model by showing actual labels and predicted ones. The model has a correct number of 745 true negatives (TN), which implies that the model has the ability to identify negative instances and the number of correct true positives (TP) is 3289 showing a very high capability to detect positive cases. In addition to these accurate predictions, it also has the 403 false positives (FP), where the negative cases are false positives, and the 203 cases are false negative (FN), where the model does not identify the actual positive cases. Those values imply that although the model in general works well, there is still some classification errors that are mainly related to mistakenly identifying negative samples as positive.

On a performance side, DistilBERT has a high level of accuracy and a high level of recall, particularly in the positive class as witnessed by high number of true positives and low number of false negatives. The fact that false positives are slightly more than false negatives also implies that the model is somewhat biased towards predicting the positive category, which is useful in the situation when it is more crucial to miss a positive example than to mistakenly claim a negative one.

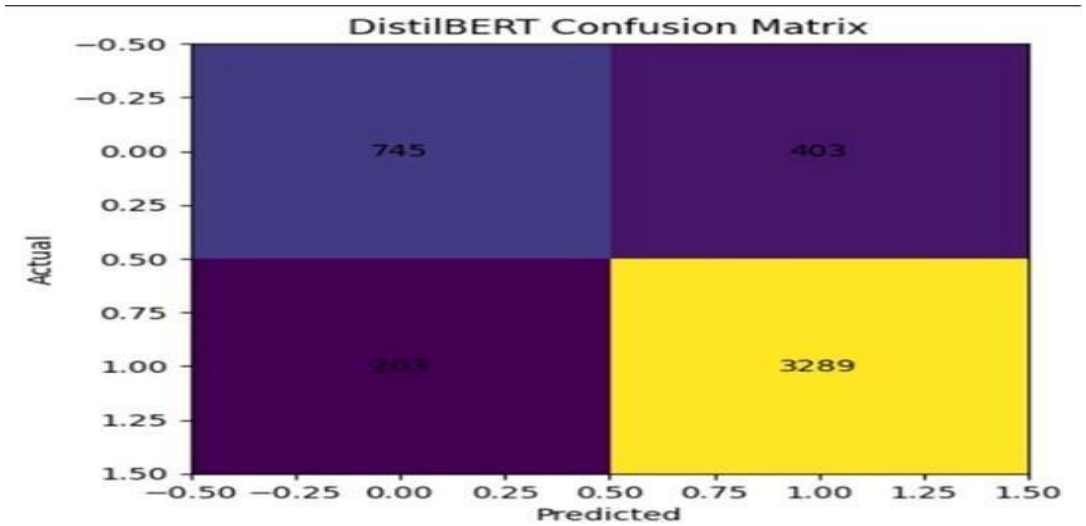


Figure 4.14: Confusion Matrix of Distil BERT Model

The Figure 4.15 presents the accuracy graph, which reveals that the DistilBERT model get more accurate with the three training epochs. The accuracy of the training increases to about 91.8 during the first epoch, and about 97.2 during the third epoch. The same tendency is noted in terms of the accuracy of the validation, which is slightly higher (approximately 92.2) and then rises to approximately 96.1. This high correlation between training accuracy and validation accuracy indicates that the model is efficiently learning and not overfitting and can generalize well to previously unseen data.

The graph of loss also supports this observation. The loss of training is much less than it was during the first epoch (around 0.20) and is only around 0.05 in the third epoch, demonstrating that the model is indeed reducing the errors. The validation loss also decreases, slowly, between about 0.08 to about 0.066 but at a slower rate than training loss. Such a small difference between training and validation loss is an indication that the model is learning effectively and at the same time that its performance is stable on validation data and that there is no serious indication of overfitting or instability.

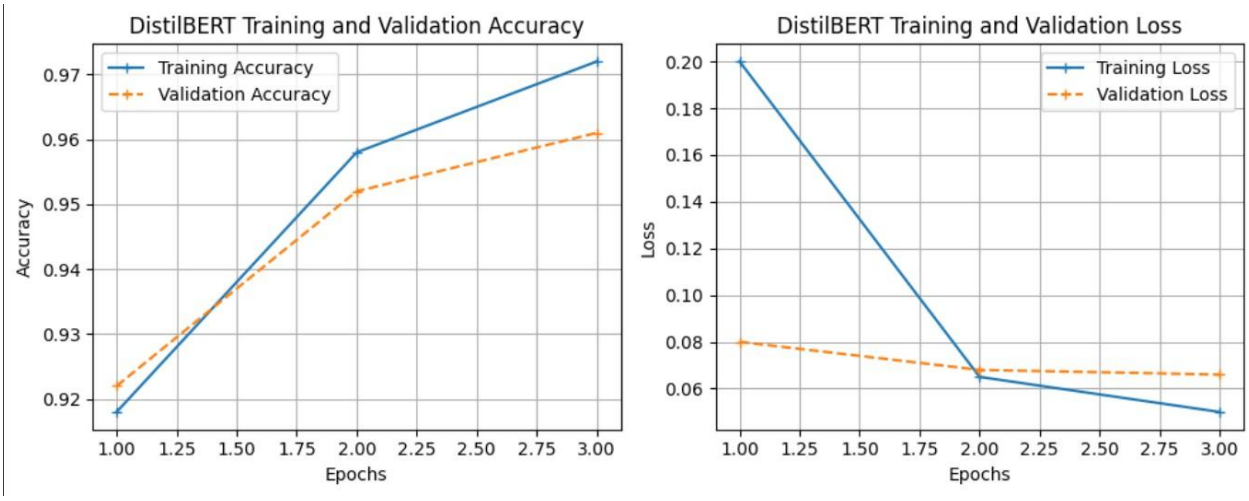


Figure 4.15: Training and Validation Curves of DistilBERT Model

4.3 Observations from the work

The Table 4.1 shows the comparison of the performances makes it clear that there is a constant improvement as more advanced models are developed with the capability of comprehending more textual data. Naive bayes model achieves the lowest accuracy (84.08) since it works under basic assumptions of probabilities and does not consider features in relationships with one another thus is not capable of venturing out the favorable language patterns. XGBoost has a much higher performance of (94.50) as it learns about interactions between the features as well as non-linear relationships, thus, it works well with standard text representations. When we go further to deep learning-based models such as LSTM (96.60%) and BiLSTM (97.00%), the accuracy gets even better since those types of models are created with the purpose to work with sequential data and obtain the contextual dependencies in the text. In particular, BiLSTM takes advantage of the possibility to process information both forward and backward and achieve a better contextual understanding.

Table 4.1. Models Performance for Fake news dataset

Models	**Accuracy(%)**	**Precision(%)**	**Recall(%)**	**F1-Score(%)**
Naïve Bayes	84.08	83.50	83.20	83.35
XGBoost	94.50	94.10	94.30	94.20
Lstm	96.60	96.20	96.50	96.35
BiLstm	97.00	96.70	96.90	96.80
Cnn+BiLstm	97.10	96.80	97.00	96.90
Bert	97.50	97.20	97.40	97.30
DistilBert	97.23	96.90	97.10	97.00

CNN+BiLSTM (97.10) model is slightly more accurate when it combines convolutional features layer with sequence learning of BiLSTM. This combination enables the model to find significant local features (like phrases) and also comprehend the context in general leading to greater accuracy and recall. Transformer models such as BERT (97.50%) and DistilBERT (97.23!) are the highest scoring models by all the metrics since they use attention mechanisms

to learn the relationship between all words in a sentence at a time. This allows them to better capture more semantic content and more subtle contextual information that can be overlooked in traditional and sequential models.

The fact that using the metrics of accuracy, precision, recall and F1-score, the values are similar in all models supports the idea that the models do not prefer to predict solely one type of background (real or fake news). The minor differences between these metrics indicate the quality of each model in terms of false positives and false negatives. In general, the findings reveal that the capability of the system to effectively and reliably identify fake news increases with both model complexity and its contextual comprehension, as does the cost of its advanced models such as BERT, making such models more appropriate to the application in practice.
Recall, precision and F1-score are more insightful in the sense that they provide an understanding of how successful the models are with respect to more than the accuracy. Recall is a measure of the model to correctly recognise all the true positive cases (finding all faked news articles). The reduced recall of Naive Bayes as a model compared to the others (83.20%) suggests that Naive Bayes fails to recall as many (in reality) fake news cases, primarily because of its simple assumptions and limited understanding of the context. As the models get more sophisticated, i. e., LSTM and BiLSTM, recall becomes highly effective (over 96 per cent) due to the ability to capture dependencies over a sequence of ensuing items and to see the context, which eliminates the possibility of overlooking important patterns. Models built on transformers as BERT, DistilBERT have the best recall since their attention components enable them to concentrate on the key aspects of the text, reduce false negatives.

Precision on the other hand indicates the number of positive cases predicted that attain the right cases. XGBoost and deep learning model approaches demonstrate high accuracy (greater than 94% much better at identifying real and fake news differences) since they are less prone to false positive results due to internal variables detecting finer differences in real/fake news. The F1-score that is a combination of both precision and recall is always high with advanced models implying that these images are at an acceptable compromise between the ability to detect fake news.

The results show a certain order of performance enhancement as the models are developed into more sophisticated deep learning and transformer-based models compared to traditional machine learning models. Naive Bayes model had a score of 84.08, which is not very high implying that it cannot identify complicated trends in textual information. The accuracy is much higher with XGBoost, at 94.50, because it is able to interact features. Replacing the transition-based model with deep learning models, LSTM achieved 96.6% accuracy, which illustrates its strength in deriving sequential relationships in text. More improvement is seen on the BiLSTM at 97.0 percent since data is processed both forward and backward, improving understanding of the context. The hybrid CNN-BiLSTM model is a little better than the others with the added benefit of feature extraction (CNN) in combination with sequence learning (BiLSTM).

Transformer-based models are most accurate at 97.5 and BERT is better at performing as it is more likely to interpret contextual relationships within the text using the mechanisms of attention. The smaller and more efficient variant of BERT is distilBERT with an extraordinarily close accuracy of 97.2, yet with a small decrease in performance, it is more efficient.

CHAPTER-5

CONCLUSION AND FUTURE WORK

In this chapter the overall work that was done in the project is summarized and the potential ways of improving it are also pointed out. The paper illustrates that Natural Language Processing (NLP), using a combination of a traditional machine learning method and the advanced deep learning approach, can be used to detect fake news with high precision. Naive Bayes, and XGBoost models offered a good baseline, with deep learning models like LSTM, Bi-LSTM, and CNN+BiLSTM demonstrating very high performance, as they could learn contextual and sequential patterns in text. In addition, models that used transformers such as BERT and DistilBERT had the best accuracies indicating that they are better at the semantic relationship of news content. Through proper text preprocessing, feature extraction, and the use of hybrid architectures, the system was able to reliably classify real and fake news articles.

5.1 Conclusion

The project was able to meet its aim of creating an effective and precise fake news detection model with a mix of machine learning, deep learning, and transformer-based solutions. The findings clearly indicate a consistent increase in the performance of the models beginning with Naive Bayes (84.08%), then XGBoost (94.50%), and subsequently deep learning models such as LSTM (96.6%) and Bi-LSTM (97.0%). The CNN+BiLSTM hybrid model with the highest accuracy of 97.1% demonstrates the benefit of integrating convolutional feature extraction with bi-directional sequence learning. BERT had the best accuracy of 97.5 and then closely came DistilBERT with a 97.2 accuracy which proves the effectiveness of transformer architecture to get deep contextual and semantic information. On the whole, the paper proves that models based on transformers are more suitable than traditional and previous deep learning models, with a high degree of generalization.

5.2 Future Study

In several ways, the current paper could be developed to enhance the quality and accuracy of the fake news detection. Future research can also aim at enhancing transformer-based models like BERT and DistilBERT by training them on bigger and more diverse data. The addition of sophisticated architectures such as RoBERTa or variations of transformers can enhance understanding of contexts further. The dataset may be increased to multilingual and cross domain news articles and this will enable easier generalization and applicability to real world. In addition, real-time stream of news analysis can be built into the system to enable it to be more viable to live detection. It is also possible to use a combination of textual data, images, videos, or metadata to implement multimodal fake news detection of the system. Lastly, it is possible to optimize the models such as DistilBERT to run faster inference to deploy the system to resource-constrained settings, without substantially affecting accuracy.

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APPENDIX

```python
from transformers import BertTokenizer, BertForSequenceClassification, Trainer, TrainingArguments

# Tokenizer
bert_tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

train_encodings = bert_tokenizer(train_texts, truncation=True, padding=True, max_length=128)
val_encodings = bert_tokenizer(val_texts, truncation=True, padding=True, max_length=128)

train_dataset = Dataset(train_encodings, train_labels)
val_dataset = Dataset(val_encodings, val_labels)

# Model
bert_model = BertForSequenceClassification.from_pretrained('bert-base-uncased', num_labels=2)

# Training args
bert_args = TrainingArguments(
    output_dir='./bert_results',
    num_train_epochs=2,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    fp16=True
)

# Trainer
bert_trainer = Trainer(
    model=bert_model,
    args=bert_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    compute_metrics=compute_metrics
)

# Train
bert_trainer.train()
```

Figure 4.16. Code snippet of Bert Model

```python
from transformers import DistilBertTokenizer, DistilBertForSequenceClassification

# Tokenizer
distil_tokenizer = DistilBertTokenizer.from_pretrained('distilbert-base-uncased')

train_encodings = distil_tokenizer(train_texts, truncation=True, padding=True, max_length=128)
val_encodings = distil_tokenizer(val_texts, truncation=True, padding=True, max_length=128)

train_dataset = Dataset(train_encodings, train_labels)
val_dataset = Dataset(val_encodings, val_labels)

# Model
distil_model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased', num_labels=2)

# Training args
distil_args = TrainingArguments(
    output_dir='./distilbert_results',
    num_train_epochs=2,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    fp16=True
)

# Trainer
distil_trainer = Trainer(
    model=distil_model,
    args=distil_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    compute_metrics=compute_metrics
)

# Train
distil_trainer.train()
```

Figure 4.17. Code snippet of DistilBert Model

```python
# LSTM model
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Embedding, Input, LSTM, Conv1D, MaxPool1D, Bidirectional, Dropout, BatchNormalization

## Creating model Using LSTM
embedding_vector_features=40
model=Sequential()
model.add(Embedding(total_words,embedding_vector_features))
model.add(Dropout(0.5))
model.add(LSTM(64))
model.add(Dropout(0.5))
model.add(Dense(1,activation='sigmoid'))
model.compile(loss='binary_crossentropy',optimizer='adam',metrics=['accuracy'])
model.build(input_shape=(None, padded_train.shape[1]))
print(model.summary())
```

Figure 4.18. Code snippet of Lstm Model

```
y_train = np.asarray(train_df['true'])
y_val = np.asarray(val_df['true'])

# Training the model
# Training the model and storing the history
history = model.fit(padded_train, y_train, batch_size = 64, validation_data=(padded_val, y_val), epochs = 3)


# Get the history data
train_accuracy = history.history['accuracy']
val_accuracy = history.history['val_accuracy']
train_loss = history.history['loss']
val_loss = history.history['val_loss']
epochs = range(1, len(train_accuracy) + 1)

# Plot Training and Validation Accuracy
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(epochs, train_accuracy, 'bo-', label='Training Accuracy')
plt.plot(epochs, val_accuracy, 'ro-', label='Validation Accuracy')
plt.title('Training and Validation Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)

# Plot Training and Validation Loss
plt.subplot(1, 2, 2)
plt.plot(epochs, train_loss, 'bo-', label='Training Loss')
plt.plot(epochs, val_loss, 'ro-', label='Validation Loss')
plt.title('Training and Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
```

Figure 4.19. Code snippet of BiLstm Model

```
# manually testing
def predict_news_cnn_bilstm(news_title):
    # Preprocess the input title
    cleaned_title = preprocess(news_title)
    # Convert to sequence and pad
    sequence = tokenizer.texts_to_sequences([cleaned_title])
    padded_sequence = pad_sequences(sequence, maxlen=42, padding='post', truncating='post')
    # Predict
    prediction = model.predict(padded_sequence)
    # Return class label
    if prediction[0] >= 0.5:
        return "True News"
    else:
        return "Fake News"

# Make a prediction
for i in range(2):
    # Get input from the user inside the loop
    user_input = input(f"Enter a news title (Prediction {i+1} of 2): ")
    result = predict_news_cnn_bilstm(user_input)
    print(f"The news title is likely: {result}")
```

Figure 4.20. Code snippet of manual testing of Cnn+BiLstm Model

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- Quoted Text
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Matches that have quotation marks, but no in-text citation

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